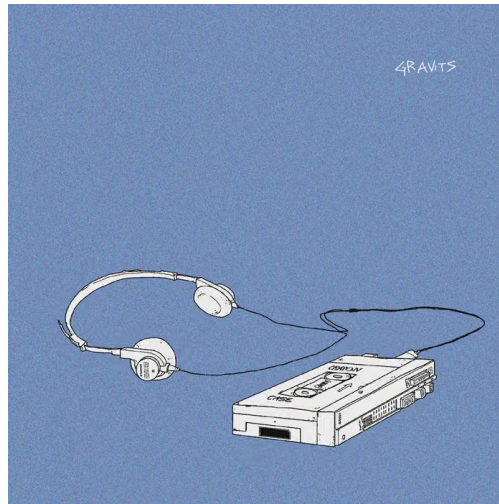


Tugas 1 IF4020 Kriptografi

Kriptanalisis pada Algoritma Kriptografi Klasik



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Pernyataan Orisinalitas

Tugas dan laporan ini disusun sepenuhnya tanpa bantuan kecerdasan buatan (*Generative AI*), melainkan hasil pemikiran dan analisis mandiri.

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Bab 1

Pendahuluan

1.1 Latar Belakang

Marisa sedang berusaha menyusun sebuah arsip rahasia yang memuat pemetaan wilayah, entitas penguasa, dan kronologi insiden penting di Gensokyo. Berkas ini merupakan gabungan dari catatan riset pribadinya serta dokumen-dokumen sensitif yang berhasil ia kumpulkan dari berbagai pihak. Demi menjaga keamanan informasi, seluruh naskah di dalam arsip tersebut dienkripsi dengan berbagai jenis *cipher* klasik. Sebagai rekan riset yang dipercaya, tugas kami adalah membantu Marisa mendekripsi serangkaian dokumen dalam arsip ini agar seluruh pengetahuan rahasia tersebut dapat terungkap.

1.2 Ringkasan Soal

Laporan ini terdiri atas lima soal wajib dan satu soal bonus. Setiap soal menuntut kriptanalisis terhadap sebuah *cipher* klasik yang berbeda.

No.	Judul	Algoritma <i>Cipher</i>
2.1	Beast Metropolis	Substitusi abjad-tunggal (<i>monoalphabetic</i>)
2.2	Horizon Chartreuse	Vigenère <i>Cipher</i>
2.3	Domain of Hopes and Dreams	Playfair <i>Cipher</i>
2.4	Trinitarian Rhapsody	Affine <i>Cipher</i>
2.5	Silent Goddess	Hill <i>Cipher</i>
3.1	Ex Machina (Bonus)	Enigma Machine M3 (termodifikasi)

Tabel 1.1: Daftar soal dan algoritma *cipher* yang digunakan.

1.3 Ketentuan Pengerjaan

Sesuai spesifikasi tugas, seluruh proses enkripsi pada *plaintext* asli dilakukan setelah teks diubah menjadi huruf kapital dan hanya pada karakter alfabetik [A-Z]; karakter lain (angka, spasi, tanda baca) dibuang dan tidak dienkripsi. Untuk soal wajib, kriptanalisis dilakukan hanya dengan *approach* yang telah diajarkan di kelas. Bantuan skrip program, lembar kerja, atau alat bantu komputasi lain diperbolehkan untuk menghitung frekuensi karakter dan pola.

Bab 2

Problem Set (Wajib)

2.1 Beast Metropolis

Deskripsi Soal

Di antara megahnya pencakar langit Alam Satwa yang dipenuhi persaingan liar, Marisa menemukan sebuah lembaran catatan yang membahas fasilitas misterius dengan struktur yang menyerupai lubang kunci. Namun, catatan mengenai benteng perlindungan tersebut kini tidak dapat dibaca karena seluruh teksnya telah dienkripsi menggunakan *cipher substitusi abjad-tunggal* (*monoalphabetic cipher*). Untuk dapat memahami struktur dari kawasan terisolasi itu, dokumen riset ini harus dikembalikan ke dalam *plaintext* seperti semula.

Bantulah Marisa mendekripsi *ciphertext* tersebut meskipun kunci pemetaannya tidak diketahui. Gunakan teknik **Analisis Frekuensi** serta analisis pola kata umum dalam **bahasa Inggris** untuk memecahkan sandi monoalfabetik ini.

Berkas lampiran:

- *Attachment:* https://drive.google.com/file/d/14DWEicqvd6_2K00cA9mbXf_ZwhPrEiak/view?usp=sharing
- md5sum: dd67736bf3c6ef6db2d5f014d55d870a
- sha256sum: a6478218a83091712dc99972553f89e92da8436b2038c5b93f782268742b4972

Berkas *Ciphertext*

```
UMRQARPQNCKICIRQQIRPVQIRZABXUASQTVUXQYQIODFDQIMZIHQSUIRDUOQUACIWQTRQAIGUOUIQJUNOXQTMFUITUCAQSCDICIVXZKQ
FDMZIDMRDFZVPCCITUFZAUCIUUAUOAMQVRZAQRPFDMZIDMRTZCDRTZYBUNUCIFCYZSUWUFBDROAQMQRZAQRPFDMZIDMZGCNURQYZB
UNUCIDFUBUNUDFUBUNUOAMQVRZAQUIKRPQFDMZIDMCTPCHZRUCCIFUIKUMZFZDFUOAMQVRZAQURRPQTUNQRCNQRPAQWQAQUXTDYQIOD
FDQIMZICIVPZSDFZAQSCDIUIKFBZTPZMDAQJUNOXQRPQUIVCQIRRDNHTDMNDRTPCCIUWUCIRPQVCRBDMUTUPCFBDRDOAMQVRZAQUIKRP
QUIVCQIRRDNHTDMRDNCFUSUCIRPQVCRBDMDFUBUNUUNDISDRPQATRPQXUASQTRDIQCTRPQRDFZRDNUUIVCQIRRDNHRPQRDRUXXQISRPMR
PQNDZIKCTUHDZRRWDPZIKAKQCSPRBNQRQATWPCVPCTNDAQRPUIRPAQRCNQTXUASQARPUIRPQXUASQTRNDZIKRDNHDZRRQCSPRBNQR
QATCIRPQXURQAQACDKMRPBUBDCOQACDKUIKRTUAQUUIKVUUVCRBUAQMUSAQURQARPUIRPQXURRQAUXRPDZSPRPQNDZIKRDNHTWCR
PUMADIRUIKHUVFVCAVZXUATPUOQWQAQUXAKUBTVURRQAQKRPADZSPDZRGUOUIKZACISRPQBUBDCOQACDKRPQQAUXBMDANTDMRPQYQIODF
DQIMZIWQAQNDZRRBEQARCVUXVUEQTRBXQTRDIQVPUNHQATWCRPHUNHDDTPUOQKWDDKQIVDMMCTWPCVPWQAQKCMMAQIRMADNRPNQNDZIKR
DNHTKZACISRPQBUBDCOQACDKNDAQDEQAQXABYQIODFDQIMZIPUKVDNNDIMQURZAQTTZVPUTUXDWQAUIKWCKQAMADIROUARVDNOUAQKRDR
PQHUVFVCAVZXUAOUARUIKRPQVDNHCIURCDIUIKODTCRCDIDMSAUEQSDDKTWAQRPQTUNQCITCKQRPQVDMMIWQAQGUQKUIKNCAADATWPCX
QDZRTCKQRPQVDMMIWQAQXUASQIZNHQADMRAUISZXUAQSKQCECIHQUTRNCADATCADIWQUODITUSACVZXZRAUXUIKMCTPCISRDDXTU
IKTDDIAQSUAKCISRPCTRUCCVPCADTPCAUCTPCHQCQEQRPURURRPURRCNQEUAACDZTMDAVQTCIFCIDXQKHBRRPBUNURUCFCISKDNUIKRPQ
TQRDCIXUIKTQUMDANQKUIUXCUIVQUIKDHURUCIQKCADUIKEUACDZTVZXZRAUXAQXCVTMADNFDAQUHBKMQMURCISMDAVQTTZVPUTIZFDFZ
```

UIKRPQCRDFCISKDNTWPCVPVDIRADXXQKRPQSQIFUCTQUDIRPQDRPQAPUIKBDTPCADFDIKDUXTDNQIRCDIQKRPURNDRDMPRQYQIODFDQIM
 ZIHZCXRDI MXRUXUIKWQAQWQXDA SUICYKUIKRPQNDZIKTWQAQXTDH ZCXR XUBQA HXBUBQAUMRQAVDITDXCKURCDIVDITCKQACISRPQT XD
 OQUIKUAUISQNIQIRDMRPQNDZIKTVCAVZXUANDZIKTWQAQVDITRAZVRQKCIUKKCRCDIHADIYQNC AADATMADNNUCIXUIKVP CUIUIKRPQWUGC
 IKQIMADNRPTUNQOQACDKCIKCVURQRPURPQAQWUTUIQJVPUI SQHQRWQKIRPQWVDZIRACQTURRPURRCNQRPQAQMDAQCRCTTOQVZXURQK
 RPURRPQQA XBUOQQAUIVQDMRPQYQIODFDQIMZINUBPUEQHQQICIMXZQIVQKHBVCECXQISCIQQACISRPQVIDXDSBMADNNUCIXUIKVP CUIU
 IKRPQODXCRCVUXUIKCKQXDCVUXCIMXZQIVQCTAQMXQVRQK CIRPQGD CIRPVNITRAZVRCDIDMRPQYQIODFDQIMZIHBEUACDZTUXXCUIVQNNQ
 NHQATUTOADMDMRPQUXXCUIVQURRPQTUNQR CNQRPQAQAQXUTDRDNHTMCEQMDZAUIKRPAQQDMRPQFUNQINZICIRPQMADIRDMRPQDPZBZUI
 RDNH C I QUTRQAIGUOUIKZACISRPQTUNQOQACDKUTCTPCHZRUCFDMZIUNDISRPQNRDNHMC EQCTRPPQUAXCQTRFDMZICIQUTRQAIGUOUIPTCT
 RDABCIKCVURCISUODTTCHXQVDIIQVRCDIHQRWQKIRPQFUYZTUOADECIVUIKRPQBUNURUCFCISKDNURRPURRCNQ CIRPQMDZARPVQIRZABP
 DFZACFZAQSCDIFUIRDAQSCDIRDUFUCAQSCDIUIKRPDPFZAQSCDIWQAQNUCIXBVDNODTQKDMADIRUIKAQUARDNHTWPCVPSAUKZUXXBHGVUN
 QYQIODFDQIMZIRWDRDRAQQSQIAURCDITXURQAASUAQKISRPCTBUNURDRUFQAZODCIRKQDZRRPURUXRPDZSPRPQDFGFCUIKICPDITPD
 FCNQIRCDIQKRPURRPQMDZASQIAUXTUIGUOUIQTQRUFQYZIUIKDRPQATPUNQASQKQUTRGUOUICIRDRPQRQAACRDBDMRPQGUOUIQTQND
 IUAVPBRPADZSPNZXRCOXQQJQKCRCDITPQHQCQEQRKRPURCWRUTUVVRZUXBFZIZIDFZICXDVRQK CIRPQIDHCOXUCIRPURMDANQKUIUXXC
 UIVQCIQUTRGUOUI NUCIXBVDITCTRCISDMAQUASAEQTCIRPQOUTRUIKRPURUMRQARPQKQURPDMPCNCFDRPQBXDTRRPQHURRXQWCRPRPQBU
 NURUCFCISKDNDAIQSDRCURQKQOVVQZIKARPPQXQKATPCODMRPQBUNURDFCISTPCOCIRPQKIRPQUXXCUIVQDMRPQKSTXUEQFCISKDNNU
 IKRPQUXXCUIVQDMRPQECXPDATQFCISKDNNAQSQK CIRDDIQMDANCISRPQGUOUIQTQNDIUAVPBRPQIRPQXQKQATDMQUTRUIKWTRGUOUIW
 QAQKCTRCISZCTPQKHBRPQYQIODFDQIMZUIKRPQMADIRUIKAQUARDNHTMADNRQQIKDMRPQMDZARPVQIRZABDRPQCMRPVQIRZABXUASQ
 YQIODFDQIMZIAQTQNHXCISDFMZICINDZIRRUCKUHQSUIRDUOQUACIOAQTQIRKUBSZINUOAQMQVRZAQXUASQYQIODFDQIMZIPUEQUXTD HQ
 QIHZCXR CIOXUVQTTZVPUBUNUIUTPCOAQMQVRZAQCHUAUF COAQMQRZAQIUIKVPCHUOAQMQVRZAQUMRQARPQNCMDZARPVQIRZABRPQKUWU
 RDNHTDACSIUXXBVQIRQAQKUADZIKRPQBUNURDFDMZICIRPQTDZRPQUTRDMRPQIUUAHUHC IUIKRPQABZHQIFDMZISAUKZUXXBTPCMRQKRD
 RPQKUCTQIFDMZICIRPQIDARPRPQUASQYQIODFDQIMZIDMKUWUWUTRPQMZA ZVPCFDMZIVXZTRQAUIKNDY ZRDNHTRPQVQIRQADMSAUECRB
 DMRPQSAQURFCISTRDNHTPCMRQKUSUCICIRPQCMRPVQIRZABDRPQUTVCQIRVCRBRDNH SADZOUIKRPQPZIKAKRDISZQKHA KRDNH SADZO
 CIRPQDTUFUOXUCIAQSUAKCISRPQAQUTDITMDARPQNZXRCOXQRAUITMGA TMRPQTQXUASQYQIODFDQIMZIHUTQKDIAQTQUAVPDIRPQFZTPC
 GCUIKICPDITPZGCRPQGUOUIQTQVUUKQNCVVDNNZICRBPUTOADODTQKECQWTTZVPUTRPQRPDABDMKBIUTRBUXRQAUIURCDIUIKRPQRPDAB
 DMQLZQTRAUIRACHQTV DILZQACISKBIUTRCQTRDQJOXUCIDIRPQDRPQAPUIKPCADTFUYZDOADODTQKRPURRPQTTRAZVRZAQTDMRPQTUTUFC
 FDGCTPCYZFCUIKNUFCIDUIVQCIRRDNH SADZOTWQAQTCNCXUAUIKADNRPPQXURQMDZARPVQIRZABDRPQTQVDIKPUXMDMRPQCMRPVQIRZA
 BRPQPQKUTDMRPQMDZAUICVQIRRDNH SADZOTGDCIRXBOXUBQKUA DXQCIRPQBUNURDFCISTPCOUMRAKQURPRPQBKQCMCQKUTRPQSUAUCUI
 KQCRBDMRPQBUNURDADBUXODWQAUIKUXTD OADODTQKRPURRPQVDITRAZVRCDIDMRPQXUASQYQIODFDQIMZIPUKUKRQAAQIRQMMQVRDIEUA
 CDZTAQSCDITURRPQTUNQR CNQ CIRPQTQVDIKPUXMDMRPQCMRPVQIRZABIDXUASQTVUXQYQIODFDQIMZIWQAQVDITRAZVRQKCIUAQUTT ZVP
 UTFCHCOADECIVQZOOQANUBBQUIKPBZSUOADECIVUIKRPQT VUXQDMRPQYQIODFDQIMZICISDFCTPCVPCKDSAU KZUXXBKQVAQTUQKUVVDAK
 CISRDRPQCITVACORCDITDIGCGCRPQCADITWDAKZIQUARPPQMADNRPPCIUWUNDZIRUCIFDMZUIKRPQXUASQTDWAKZIQUARPPQMADNRPPQR
 UMZIU BUNUFDMZIQNOQADABZABUFZURRPURRCNQAQSUAKQKPCNTQXMTRPQFCISDMRPQWDAKXUIKPTUZRPD ACBWTATAQMXQVRQK CIRPQMU
 VRRPURDIXBPQVDIRCIZQRDHCXKXUASQYQIODFDQIMZICIRPQGUOUIQTQUAVPCOQXUSD CIRPQTCJRPVQIRZABRPQPDACYDIRUXVUEQTRB
 XQTRDIQVPUNHQADMRPQFCIUCTBTQRNHQSUIRDHQDOZXYUACQKCIYQIODFDQIMZIUURRPQTUNQR CNQRPQBHSUIRDUOQUACIUAQUTT ZVP
 TTQRTZOAD E C I V Q D W U A C O A D E C I V Q Z G C O A D E C I V Q I K Q E Q I C I F U I R D A Q S C D I W P C V P W Q A Q V D I T C K Q A Q K U T H U T Q T M D A R P Q Q T R U H X C T P N Q I R D M
 RPQQNOQADAFQCRUCUNDISRPQNZQIDPUKUOUARCVZXUAXBXUASQIZNHQADMYQIODFDQIMZIVDNOUAQKRDDIXBRPCARBIQICIFDYZFQOADE
 CIVQKZACISRPQTUNQOQACDKZQIDUXDIQPUKICIQRBTEQIWPXCQIFUIRDRPQAQWQAQRWDPZIKAKTCTJRPVQIUTHDRPFCIDUIKDCYPUISKQ
 VAQUTQKDI XBRPQFUIRDAQSCDIDIRCIZQRDVIDITRAZVRXUASQYQIODFDQIMZIRPADZSPDZRRPQTCJRPVQIRZABCIKCVURCISRPURRPQBU
 NURDNDIUAVPBRRPURRCNQAQXCQKQKUECXBDIRPQUTRQAIFCISKDNCIHDPQVDIDNCVUIKNCXCRUABUTOQVRTMADNRPPQIKDMRPQTCJRP
 VQIRZABDRPQHQSICICISDMRPQTQE QIRPVQIRZABXUASQTLZUAQIUKVCAVZXUARDNHTAQOXUVQKRPQMADIRUIKHQVUNQRPNUCITRAQUNC
 IRPNCKKXQDMRPQTQE QIRPVQIRZABRPQBQEDXEQK CIRDDVRUSDIXRDNHTAQSUAKCISRPQAQUTDIMDARPQVQTTURCIDMVDITRAZVRCDID
 MYQIODFDQIMZIRUCCVPCADTPCAUCTPCHQCQEQTRPURCRCTAQXURQKDRPQAQMDANTDMVADWIOACIVQTPDRDFZUIKTD SUIDZNUF DUTWQXX
 UTRPQQTRUHXCTPNQIRDMRPQIURCDIUXNUIZMUVRZACISTBTRQNWPCXQRPQMCIXUYQIODFDQIMZIU AQHQCQEQRD HQUTUNUNDZIRUCIFDM
 ZIUIKADISNUVPCVXCNTRPURRPQFDMZIWQAQHZCXRCIRPQMCATRPXMDMRPQTQE QIRPVQIRZAB

Langkah Penyelesaian

Monalphabetic cipher rentan terhadap analisis frekuensi. Jika kriptanalisis mendapatkan frekuensi yang mendekati distribusi huruf pada kata-kata di Bahasa Inggris, sebuah *ciphertext* dapat diterka dengan mencocokkan monogram, bigram, bahkan hingga trigram.

Langkah-langkah yang dilakukan:

1. Lakukan penghitungan frekuensi monogram pada *ciphertext*.

```

uv run 21/solver.py
rank cipher cipher% -> plain plain%
1 Q 12.31 -> E 12.02
2 R 9.27 -> T 9.10
3 U 8.83 -> A 8.12
4 I 8.60 -> O 7.68
5 D 8.06 -> I 7.31
6 C 6.64 -> N 6.95
7 A 6.10 -> S 6.28
8 P 5.61 -> R 6.02
9 T 4.30 -> H 5.92
10 K 3.56 -> D 4.32
11 Z 3.56 -> L 3.98
12 M 3.07 -> U 2.88
13 V 2.89 -> C 2.71
14 X 2.81 -> M 2.61
15 N 2.73 -> F 2.30
16 S 2.01 -> Y 2.11
17 O 1.96 -> W 2.09
18 B 1.78 -> G 2.03
19 F 1.77 -> P 1.82
20 H 1.34 -> B 1.49
21 W 0.96 -> V 1.11
22 E 0.64 -> K 0.69
23 Y 0.59 -> X 0.17
24 G 0.39 -> Q 0.11
25 J 0.15 -> J 0.10
26 L 0.05 -> Z 0.07

```

Kita dapatkan pemetaan (kasar) berikut:

cipher	ABCDEFGHIJKLMNOPQRSTUVWXYZ
plain	SGNIKPQBOJDZUFWRETYHACVMXL

2. Untuk memperkuat dasar pencocokan, hitung juga frekuensi bigram dan trigram pada *ciphertext* (`freq_ngram(ciphertext, n)` dengan $n = 1, 2, 3$), lalu bandingkan sepuluh besarnya dengan referensi Bahasa Inggris.

Monogram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	Q	12.31	E	12.02
2	R	9.27	T	9.10
3	U	8.83	A	8.12
4	I	8.60	O	7.68
5	D	8.06	I	7.31
6	C	6.64	N	6.95
7	A	6.10	S	6.28
8	P	5.61	R	6.02
9	T	4.30	H	5.92
10	K	3.56	D	4.32

Bigram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	RP	3.88	TH	3.56
2	PQ	2.99	HE	3.07
3	CI	2.21	IN	2.43
4	UI	1.83	ER	2.05
5	QI	1.80	AN	1.99
6	AQ	1.70	RE	1.85
7	IR	1.64	ON	1.76
8	QA	1.49	AT	1.49
9	DI	1.42	EN	1.45
10	IK	1.39	ND	1.35

Trigram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	RPQ	2.91	THE	1.81
2	UIK	1.03	AND	0.73
3	QIR	0.62	ING	0.72
4	MZI	0.62	ENT	0.42
5	DMR	0.61	ION	0.42
6	MRP	0.61	HER	0.36
7	IRP	0.59	FOR	0.34
8	CIS	0.56	THA	0.33
9	QAQ	0.52	NTH	0.33
10	KRP	0.51	INT	0.32

Bentuk distribusinya memang tidak identik urutan-per-urutan, tetapi tiga huruf/pasangan/tripel tersering pada *ciphertext* (Q, R, U dan RP, PQ dan RPQ) menempati rentang persentase yang sebanding dengan huruf/pasangan/tripel tersering Bahasa Inggris (E, T, A dan TH, HE dan THE). Hal ini konsisten dengan hipotesis awal bahwa *cipher*-nya adalah substitusi abjad-tunggal murni terhadap teks Bahasa Inggris, sehingga pencocokan berbasis frekuensi dapat dilanjutkan.

3. Lakukan substitusi berdasarkan pemetaan (kasar) yang telah dibuat.

AUTESTREFNDONOEETOTRCEOTLSGMASYEHKAMEXEOWIPIEOULOBEOAOTIAWWEASNOVEHTESQAWAOEJAFWMEHIUPAOHANSEYNIONO
CMLDEPIULOIUTIPLCRNNOHAPLSANOASAWSEUECTLSETREPIULOIUTHNLTHLXLGAFANOPNXYAVAPGITIWSEUECTLSETREPIULOIUI
LQNFATEXLGAFANOIPAGAFAPAGAFWSEUECTLSEAODTREPIULOIUNHRLTANNOPAOAULPLIPAWSEUECTLSEATTREHAFETNFETRE
SEVESEAMHIXEOWIPIEOULONOCRLIPLSEYNIOAODPGLHRLUISEJAFWMETREAOACNEOTTIFBHIUFIITHRNOAVANOTRECNIGUAHARN
PGITIWSEUECTLSEAODTREAOACNEOTTIFBHIUFIIFNPAYANOTRECNIGUIPAGAFAAFIOTITRESHTREMASYEHTIOENHTRETIFLIFAAOC
NEOTTIFBTRETITAMMEOTRIUTREFILODNHABILTTVIRLODSEDNRYRTGFETESHVRNCRNHFISETRAOTRSEETNFEHMASYESTRAOTREMA
SYEHTFILODTIFBUIABILTENYRTGFETESHNOTREMATESWESNIDIUTREGAGINWESNIDAODNTHASEAAODCAWACNTGASEUASYSEATESTR
AOTREMATTESAMTRILYRTREFILODTIFBHVNTAUSIOTAODBACPCNSCLMASHRAWVESEAMSEADGHCATTESEDTRSILYRILTQAWAODLSN
OYTREGAGINWESNIDTREEASMGUISFHIUTREXEOWIPIEOULOVESEFIHTMGKESTNCAMCAKEHTGMEHTIOECRAFESHVNTREBAFBIHRAW
DVIIDEOCIUUNOHVRNCRVESEDNUVESEOTUSIFTREFILODTIFBHDLSNOYTREGAGINWESNIDFISEIKESEASMGXEOWIPIEOULORADCIFF
IOUEATLSEHHL CRAHAMIVESAODVNDUSIOTWASTCWFASEDTITREBACPCNSCLMASWASTAODTRECIFBNATNIOADWIHNTNIOIUYSA
KEYIIDHVESETREHAFENOHNDRECIUUNOVESEQADEAODFNSSISHVRNMEILTHNDETRECIUUNOVESEAMASYEOLFBESIUTSNAOYLMASE
DYEDKNNOEBEAHTFNSSISHNSIOVEAWIOHAYSNCMLTMSAMAODUNHRNOYTIIMHAODHIIIOSEYASDNOYTRNHTANNCNRNSIHRNSANHRNBEMN
EKEDTRATATTRATTFEKASNILHUISCEHNOPNOIMEDBGTRREGAFATANPNYDIFAODTREHETINOMADHEAUISEFEDAOAMNNOCEAODIBTA
NOEDNSIOADKASNILHCLMTLSAMSEMNCUSIFPISEABGDEUEATNOYUISCEHHL CRAHOLPIPLAODTRENTIPNOYDIFHVRNCRCIOTSIMME
DTREYEOPANHEATOTREITRESAODGIHRNSIPIODIAMHIFEOTNIOEDTRATFIHTIUTREXEOWIPIEOULOBMLNMTIOMATMAODVESEVEMMI
SYAONXEDAODTREFILODHVESEAMHIBLNM TMAGESBGMAGESAUTESCIOHIMNDATNIOCIHNDENSOYTREHMIWEAODASSAOYEFEOIUTRE
FILODH CNSCLMASFILODHVESECIOHTSLCTEDNOADDNTNIOBSIOXEFNSSISHUSIFFANOMAODCRNOAAODTREVAQNODEOUSIFTREHAFEW
ESNIDNODNCATETRATTRESEVAHAOEJCRAOYEBETVEEOTRETVICILOTSNEHATTRATTFETRESEUISENTNHHWECLMATEDTRATTREEASM
GAWWEASAOCEIUTREXEOWIPIEOULOFAGRAKEBEEONUOMLEOCEDBGCKNMEOYNOEESNOYTECROIMIYGUSIFFANOMAODCRNOAAODTREW
IMNTNCAMAODNDEIMINYCAMNOUMLEOCENHSEUMECTEDNOTREQINOTCIOHTSLCTNIOIUTREXEOWIPIEOULOBGKASNILHAMNNOCEFEF
BESHAWHSIIUIUTREAMNNOCEATTREHAFETNFETRESEASEAMHITIFBHNKUEILSAODTRSEEIUTREPAFEOLFONOTREUSIOTIUTRERIL

GLAOTIFBNOEAHTESQAWAODLSNOYTREHAFEWESNIDAHNRNBLTANPIULOAFIOYTREFTIFBUNKENHTREEASMNEHTPIULONOEAHTESO
 QAWAOHRNHTISGNODNCATNOYAWIHHNBMECIOECTNIOBETVEOTREPAXLHAWSIKNOCEAODTREGAFATANPNODYIFATTRATTNFENOTRE
 UILSTRCEOTLSGRIPLSNPLSEYNIOPAOTISEYNIOTIPANSEYNI OAODTIRIPLSEYNI OVESEFANOMGCIFWIHEDIUUSIOTAODSEASTIFBH
 VRNCRYSADLAMMBECAFEXEOWIPIEOULOTVITITRSEEYEOESATNIOHMATESSEYASDNOYTRNHGAFATITAPESLWINOTEDILTTRATAMTR
 ILYRTREPIQNPNAODONRIOHRIPNFEOTNIOEDTRATTREUILSYEOESAMHAODQAWAOEHETAPEXLOAODITRESHRADFESYEDEAHTQAWAONO
 TITRETESNTISGIUTREQAWAOEHEFIOASCRGTRSILYRFLMTNWMEJWEDNTNIOHREBEMNEKEDTRATNTVAHACTLAMMGLOLOIPLONMIC
 ATEDNOTREOIBNWMANOTRATUISFEDAOAMMNAOCENOEAHTQAWAOFANOMGCIOHNHTNOYIUSEASYSAKEHNOTREWAHTAODTRATAUTESTRE
 DEATRIURNFPITREGMIHTTREBATTMEVNTRTREGAFATANPNODYIFISOEYITNATEDWEACELODESTREMEADESHRNWIUTREGAFATIPNOY
 HRNWNOTREODTREAMMNAOCEIUTREDIYHMAKEPNOYDIFAODTREAMMNAOCEIUTREEKNMRISHEPNOYDIFESYEDNOTIOEUISFNOYTRE
 QAWAOEHEFIOASCRGTREOTREMEADESHIUEAHTAODVEHTQAWAOVESEDNHTNOYLNHREDBGTREXEOWIPIEOULO AODTREUSIOTAODSEAST
 IFBHUSIFTREEODIUTREUILSTRCEOTLSGTITREUNTRCEOTLSGMASYEXEOWIPIEOULOSEHEFBMNOYPIULONOFILOTTANDABEYAOITIA
 WWEASNOWSEHEOTDAGYLOFAWSEUECTLSEMASYEXEOWIPIEOULORAKEAMHIBEEOBLNMTNOWMACEHLCRAHGAF AOAHRNWSEUECTLSENB
 ASAPNWSEUECTLSEAODCRNBWSEUECTLSEAUTESTREFNDUILSTRCEOTLSGTREDAVATIFBHSISNYNOAMMGCEOTESEDASILODTREGAFAT
 IPULONOTREHILTREAHITUTREOASABAHNOAODTRESGLBEOPULYOYADLAMMGHRNUTEDTITREDANHEOPIULONOTREOISTRTREMASYE
 XEOWIPIEOULOIUDAVAVAHTREULSLNCRNPIULOCMLHTESAODFXILTFIBHTRECEOTESIUYSAKNTGIUTREYSEATPNODYITFBNRUTEDA
 YANONOTREUNTRCEOTLSGTITREAOCEOTCNTGTIFBYSILWAODTRERLODSEDTIOYLEDNBSDTIFBYSILWNOTREIHAPAWMANOSEYASDN
 OYTRESEAHIOHUISTREFLMTNWMETSAOHUESHIUTREHEMASYEXEOWIPIEOULOBABEDIOSEHEASCRITOTREPLHRNQNAODONRIOHRLQNTR
 EQAWAOEHEACADEFNCCIFFLONTGRAHWSIWIHEDKNEVHHL CRAHTRETREISGIUDGOAHTGAMTESOATNIOAODTRETREISGIEZLEHTSAOT
 SNBEHCIOZLESNOYDGOAHTNEHTIEJWMANOIOTREITRESRAODRNSIHEPAXLWSIWIHEDTRATTREHTSLCTLSEHIUTREHAHAPNPIQHNRN
 XLPAODFAPNOIAOCNEOTTIFBYSILWHVESEHNFNMASAOUSIFTREMAUEUILSTRCEOTLSGTITREHECIODRAMUIUTREUNTRCEOTLSGT
 REREDHIUTREUILSAOCNEOTTIFBYSILWHQINOTMGWMAGEDASIMENOTREGAFATIPNOYHRNWAUTESDEATRREGDENUNEDAHTREYLASD
 NAODENTGIUTREGAFATISIGAMWIVESAODAMHIWSIWIHEDTRATTRECIOTSLCTNIOIUTREMASYEXEOWIPIEOULORADADETSESOTEUU
 ECTIOKASNILHSEYNIOHATTREHAFETNFENOTREHECIODRAMUIUTREUNTRCEOTLSGOIMASYEHKAMEXEOWIPIEOULOVESECIOHTSLCT
 EDNOASEAHHL CRAHPNBNWSIKNOCELWESFAIGEAO DRGLYAWSIKNOCEAODTREHCA MEIUTREXEOWIPIEOULONOIYIPNHRNCRNDIYSADLA
 MMGDECSEAHEDACCISDNOYTTIRENOHCSNWTNIOHQNTRENSIOHVISDLOEASTREDUSIFTRENOAVAFILOATANOPULOAODTREMAYE
 HVISDLOEASTREDUSIFTREETAULOAGAFAPIULOEFWESISGLSGAPLATTRATTNFSEYASDEDRNFHEMUAHTREPNOYIUTREVISMADOARNH
 ALTRISNTGVAHSEUMECTEDNOTREUACTTRATIOMGRECIOTNOLEDITBLNMDMASYEXEOWIPIEOULONOTREQAWAOEHEASCRNWEMAYINOTR
 EHNJTRCEOTLSGTRERISNXIOTAMCAKEHTGMEHTIOECRAFBSIUTREPNOANHGHTEFBEYAOITIBEWIWLMSXEDNOXEOWIPIEOULOATTR
 EHAFETNFETREGBEYAOITIAWWEASNOASEAHHL CRAHHETTHLSIKNOCEIVASNWSIKNOCELQNWSIKNOCEAODEKEONOPAOITISEYNIORVNC
 RVESECIOHNDSEDAHBAHEHUISTREEHTABMNRHFEOTIUTREEFWESISPENTANAFIOYTREFLEOIRADAWASTNCLMASMGMASYEO LFBESI
 XEOWIPIEOULOCIFWASEDTIOMGTRNSTGONENOPIXLPEWSIKNOCEDLSNOYTREHAFEWESNIDLEOIAMIOERADONOTGHEKEOVRNMENO
 PAOTITRESEVESETVIRLODSEDHNTJTEOAHBITRPNIOAODINXRAOYDECSEAHEDIOGTREPAOTISEYNIOCIOTNOLEDTICIOHTSLCTMAS
 YEXEOWIPIEOULOTRSILYRILTTREHNJTRCEOTLSGNODNCATNOYTRATTREGAFATIFIOASCRGATTRATTNFSEMNEDREAKNMGIOTREAH
 TESOPNOYDIFNOBITRECIOIFNCAODFMNTASGAHWECETHUSIFTREEODIUTREHNJTRCEOTLSGTITREBEYNOONOIYUTREHEKEOTRCEOTL
 SGMASYEHZLASEAODCNSCLMASTIFBHSEWMACEDTREUSIOTAODBECAFETREFANOHTSEAFNOTREFNDMEIUTREHEKEOTRCEOTLSGTREG
 EKIMKEDNOTIICTAYIOAMTIFBHSEYASDNOYTRESEAHIOUISTRECEHATNIOIUCIOHTSLCTNIOIUXEOWIPIEOULOTANNCRNSIHRNSAN
 HRNBEMNEKEHTRATNTNHSEMATEDTITRESEUISFHIUCSIVOWSNOCEHRITIPLAODHIYA OILFAPIAHVEMMAHTREEHTABMNRHFEOTIUTRE
 OATNIOAMFAOLUACTLSNOYHGHTFVRNMETREUNOAMXEOWIPIEOULOASEBEMNEKEDTIBEAHAFAFILOATANOPULOAODSIOYFACRNCMAN
 FHTRATTREPIULOVESEBLNMTNOTREUNSHTRAMUIUTREHEKEOTRCEOTLSG

4. Cocokkan melalui informasi kecocokan bigram dan trigram, lalu lanjutkan dengan iterasi yang sama: setiap kali sebuah bigram/trigram/kata umum Bahasa Inggris mulai terlihat pada hasil dekripsi semantara, huruf *cipher* yang bersangkutan diganti agar sesuai. Tiga huruf – R, Q, U – sudah benar sejak tebakan awal (langkah 1) dan tidak pernah diubah lagi. Untuk sisanya, dapat dilihat pada tabel di bawah yang disusun berdasarkan *log* dari *src/21/log/plain/* secara kronologis.

Ganti	Konteks (sebelum → sesudah)	Petunjuk
P→H	AUTESTREFNDONOETEEOTR → AUTESTHEFNDONOETEEOTH	delapan karakter pertama UMRQAR PQ → AUTESTHE, trigram THE
I→N	UTESTHEFNDONOETEEOTHCEOTL UTESTHEFNDNNNTEENTHCENTL →	mengarah ke NINETEENTH
C→I	AUTESTHEFNDNNNTEENTHCEN AUTESTHEFIDNINETEENTHCEN →	kata NINETEENTH terbentuk
A→R	AUTESTHEFIDNINETEEN → AUTERTHEFIDNINETEEN	bigram ER
Z→U	TEENTHCENTLSGMASYEHCAMEXE TEENTHCENTURYMARYEHCAMEXE →	mengarah ke CENTURY
B→Y	ENTHCENTLSGMASYEHCAMEXENW ENTHCENTURYMARYEHCAMEXENW →	kata CENTURY lengkap
M→F	AUTERTHEFIDNINET → AFTERTHEFIDNINET	kata umum AFTER THE
N→M	AFTERTHEFIDNINETEENTHCE AFTERTHEMIDNINETEENTHCE →	kata MID (AFTER THE MID-...)
D→O	EHCAMEXENPIPIENFUNBEYANTI EHCAMEXENPOPOENFUNBEYANTO →	mengarah ke ZENPOKOENFUN
T→S	NTURYMARSEHCAMEXENPOPOENF NTURYMARYESCAMEXENPOPOENF →	bigram ES
F→K	SCAPEXENPOPOENFUNBEYANTOA SCAPEXENPOKOENFUNBEYANTOA →	mengarah ke ZENPOKOEN
S→G	CENTURYPARYESCAPEXENPOKOE CENTURYPARGESCAPEXENPOKOE →	kata LARGE
O→P	GESCAPEXENLOKOENFUNBEGANT GESCAPEXENPOKOENFUNBEGANT →	kata ZENPOKOEN
X→L	NTHCENTURYPARGESCAPEXENPO NTHCENTURYLARGESCALEXENPO →	kata LARGE SCALE
W→W	TOAPPEARINGESTERNQAPANEJA TOAPPEARINWESTERNJAPANEXA →	kata IN WESTERN
G→J	RINGESTERNQAPANEJAMPLESOF RINWESTERNJAPANEXAMPLESOF →	kata JAPAN
J→X	TERNQAPANEJAMPLESOFKANSAI TERNJAPANEXAMPLESOFKANSAI →	kata EXAMPLES
Y→Z	LARGESCALEXENPOKOENFUNBEG LARGESCALEZENPOKOENFUNBEG →	istilah ZENPOKOENFUN terbentuk
E→V	WEREMOSTLYKERTICALCAKESTY WEREMOSTLYVERTICALCAVESTY →	kata VERTICAL CAVE
L→Q	LARGESZUAREANDCIR → LARGESQUAREANDCIR	kata SQUARE

Artefak

Skrip *solver* yang digunakan untuk menghitung frekuensi monogram/bigram/trigram, menyusun tebakan awal, menerapkan pemetaan hasil analisis, serta mencatat setiap hasil dekripsi ke log/plain/ dan metadatanya (pemetaan & frekuensi n-gram) ke log/metadata/. Modul ini bergantung pada `src/frequencies.py` yang berisi tabel referensi frekuensi Bahasa Inggris (dipakai bersama untuk seluruh soal substitusi/Vigenère pada laporan ini, sehingga isinya tidak diduplikasi di sini).

```

1 import json
2 import os
3 import sys
4 from datetime import datetime
5 from typing import Dict, List
6
7 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
8 from frequencies import MONOGRAM, MONOGRAM_ORDER
9
10 CORRECTIONS = [
11     ("R", "T"), ("P", "H"), ("Q", "E"), ("I", "N"), ("C", "I"),
12     ("U", "A"), ("Z", "U"), ("A", "R"), ("B", "Y"), ("M", "F"),
13     ("O", "P"), ("W", "W"), ("N", "M"), ("D", "O"), ("T", "S"),
14     ("X", "L"), ("G", "J"), ("F", "K"), ("S", "G"), ("J", "X"),
15     ("Y", "Z"), ("E", "V"), ("L", "Q"),
16 ]
17

```

```

18 class Solver:
19     def count(self, string: str) -> Dict:
20         data = {}
21         for c in string:
22             data[c] = data.get(c, 0) + 1
23         return dict(sorted(data.items(), key=lambda item: item[1], reverse=True))
24
25     def sort(self, string: str) -> List:
26         return list(self.count(string).keys())
27
28     def freq_ngram(self, string: str, n: int = 1) -> Dict:
29         freq_data = {}
30         total_ngrams = len(string) - n + 1
31         for i in range(total_ngrams):
32             ngram = string[i:i+n]
33             freq_data[ngram] = (freq_data.get(ngram, 0) + 1)
34         for ngram in freq_data:
35             freq_data[ngram] = (freq_data[ngram] / total_ngrams * 100)
36         return dict(sorted(freq_data.items(), key=lambda item: item[1], reverse=
37 True))
38
39     def solve(self, string: str) -> Dict:
40         cipher_order = [c for c in self.sort(string) if c.isalpha()]
41         mapping = {}
42         for cipher_letter, plain_letter in zip(cipher_order, MONOGRAM_ORDER):
43             mapping[cipher_letter] = plain_letter
44         return mapping
45
46     def replace_char(self, mapping: Dict, char0: str, char1: str) -> Dict:
47         mapping[char0] = char1
48         return mapping
49
50     def decrypt(self, string: str, mapping: Dict) -> str:
51         return "".join(mapping.get(c, c.lower()) for c in string)
52
53     def table(self, string: str) -> str:
54         cipher_freq = self.freq_ngram(string)
55         rows = ["rank cipher cipher% -> plain plain%"]
56         for i, cipher_letter in enumerate(cipher_freq):
57             if not cipher_letter.isalpha():
58                 continue
59             plain_letter = MONOGRAM_ORDER[i] if i < len(MONOGRAM_ORDER) else "-"
60             plain_pct = MONOGRAM.get(plain_letter, 0.0)
61             rows.append(
62                 f"{i+1:>4} {cipher_letter:^6} {cipher_freq[cipher_letter]:>6.2f}
63                 -> "
64                 f"{plain_letter:^5} {plain_pct:>5.2f}"
65             )
66         return "\n".join(rows)
67
68     def load_ciphertext(self, base_dir: str) -> str:
69         with open(os.path.join(base_dir, "s.txt")) as f:
70             return f.read().strip()
71
72     def write_plain(self, base_dir: str, timestamp: str, decrypted: str) -> str:
73         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
74         os.makedirs(os.path.dirname(path), exist_ok=True)
75         with open(path, "w") as f:
76             f.write(decrypted)

```

```

75     return path
76
77 def write_metadata(base_dir: str, timestamp: str, mapping: Dict, freq: Dict) -> str
78 :
79     path = os.path.join(base_dir, "log", "metadata", timestamp + ".json")
80     os.makedirs(os.path.dirname(path), exist_ok=True)
81     payload = {
82         "timestamp": datetime.now().isoformat(timespec="seconds"),
83         "mapping": mapping,
84         "freq_ngram": {str(n): freq[n] for n in freq},
85     }
86     with open(path, "w") as f:
87         json.dump(payload, f, indent=2)
88     return path
89
90 if __name__ == "__main__":
91     base_dir = os.path.dirname(os.path.abspath(__file__))
92     content = load_ciphertext(base_dir)
93
94     solver = Solver()
95     mapping = solver.solve(content)
96     for cipher_letter, plain_letter in CORRECTIONS:
97         mapping = solver.replace_char(mapping, cipher_letter, plain_letter)
98
99     decrypted = solver.decrypt(content, mapping)
100     freq = {n: solver.freq_ngram(content, n) for n in (1, 2, 3)}
101
102     timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
103     plain_path = write_plain(base_dir, timestamp, decrypted)
104     metadata_path = write_metadata(base_dir, timestamp, mapping, freq)
105
106     print("mapping (cipher -> plain):")
107     print("".join(sorted(mapping)))
108     print("".join(mapping[c] for c in sorted(mapping)))
109     print()
110     print("hasil dekripsi:")
111     print(decrypted)
112     print()
113     print(f"[tersimpan ke {os.path.relpath(plain_path)} dan {os.path.relpath(
114         metadata_path)}]")

```

src/21/solver.py

Plaintext Hasil Dekripsi

AFTER THE MIDNINETEENTH CENTURY LARGE SCALE ZENPO KOEN FUN BEGAN TO APPEAR IN WESTERN JAPAN. EXAMPLES OF KANSAI REGION INCLUDE KOFUN OF TOKUCHI IN SAKURAI NARA PREFECTURE, THE KOFUN OF TSUJITSU, YAMAIKIZUGAWA KYOTO PREFECTURE, THE KOFUN OF FUJIMATEZUYAMA IN OKAYAMA OKAYAMA PREFECTURE AND THE KOFUN OF ISHIBUTAI IN KANDAFUKUOKA PREFECTURE. AT THE SAME TIME THERE WERE ALSO ZENPO KOEN FUN IN CHUGOKU REGION AND KYUSHU. FOR EXAMPLE THE ANCIENT TOMBS OF MOTOSHI IN NAWA IN THE CITY OF ASAHIKYOTO PREFECTURE AND THE ANCIENT TOMBS OF TOMIKAGA IN THE CITY OF OKAYAMA AMONG OTHERS. THE LARGEST ONE IS THE TOKUTOMA ANCIENT TOMB. THE TOTAL LENGTH OF THE MOUND IS ABOUT TWO HUNDRED EIGHTY METERS, WHICH IS MORE THAN THREE TIMES LARGER THAN THE LARGEST MOUND TOMB OF ABOUT EIGHTY METERS IN THE LATER PERIOD OF THE YAYOI PERIOD AND ITS AREA AND CAPACITY ARE FAR GREATER THAN THE LATTER. ALTHOUGH THE MOUND TOMBS WITH A FRONT AND BACK CIRCULAR SHAPE WERE ALREADY SCATTERED THROUGHOUT JAPAN DURING THE YAYOI PERIOD, THE EARLY FORMS OF THE ZENPO KOEN FUN WERE MOSTLY VERTICAL CAVE STYLE STONE CHAMBERS WITH BAMBOO SHAPED WOODEN COFFINS WHICH WERE DIFFERENT FROM THE MOUND TOMBS DURING THE YAYOI PERIOD. MORE OVER EARLY ZENPO KOEN FUN HAD COMMON FEATURES SUCH AS A LOWER AND WIDER FRONT PART COMPARED TO THE BACK CIRCULAR PART AND THE COMBINATION AND POSITION OF GRAVE GOODS WERE THE SAME INSIDE THE COFFIN. THERE WERE JADE AND MIRRORS WHILE

EOUTSIDETHECOFFINWEREALARGENUMBEROFTRIANGULAREGDEDIVINEBEASTMIRRORSIRONWEAPONSAGRICULTURALANDFISHINGTOOLSANDSOONREGARDINGTHISAIICHIROSHIRAIISHIBELIEVEDTHATATTHATTIMEVARIOUSFORCESINKINOLEDBYTHEYAMATAIKINGDOMANDTHESETOINLANDSEAFORMEDANALLIANCEANDOBTAINEDIRONANDVARIOUSCULTURALRELICSFROMKOREABYDEFEATINGFORCESSUCHASNOKUKUANDTHEITOKINGDOMSWHICHCONTROLLEDTHEGENKAISEAONTHEOTHERHANDYOSHIROKONDOALSOMENTIONEDTHATMOSTOFTHEZENPOKOENFUNBUILT ON FLATLAND WERE WELL ORGANIZED AND THE MOUNDS WERE ALSO BUILT LAYER BY LAYER AFTER CONSOLIDATION CONSIDERING THE SLOPE AND ARRANGEMENT OF THE MOUNDS CIRCULAR MOUNDS WERE CONSTRUCTED IN ADDITION BRONZE MIRRORS FROM MAINLAND CHINA AND THE WAJINDEN FROM THE SAME PERIOD INDICATE THAT THERE WAS AN EXCHANGE BETWEEN THE TWO COUNTRIES AT THAT TIME THEREFORE IT IS SPECULATED THAT THE EARLY APPEARANCE OF THE ZENPOKOEN FUN MAY HAVE BEEN INFLUENCED BY CIVIL ENGINEERING TECHNOLOGY FROM MAINLAND CHINA AND THE POLITICAL AND IDEOLOGICAL INFLUENCE IS REFLECTED IN THE JOINT CONSTRUCTION OF THE ZENPOKOEN FUN BY VARIOUS ALLIANCE MEMBERS AS PROOF OF THE ALLIANCE AT THE SAME TIME THERE ARE ALSO TOMBS FIVE FOUR AND THREE OF THE KAMENMUN IN THE FRONT OF THE HOUYUAN TOMB IN EASTERN JAPAN DURING THE SAME PERIOD AS ISHIBUTAI KOFUN AMONG THE TOMB FIVE IS THE EARLIEST KOFUN IN EASTERN JAPAN HIS TORY INDICATING A POSSIBLE CONNECTION BETWEEN THE KAZUS A PROVINCE AND THE YAMATAI KINGDOM AT THAT TIME IN THE FOURTH CENTURY HOKURIKU REGION KANTO REGION TOKAI REGION AND TOHOKU REGION WERE MAINLY COMPOSED OF FRONT AND REAR TOMBS WHICH GRADUALLY BECAME ZENPOKOEN FUN TWO TO THREE GENERATIONS LATER REGARDING THIS YAMATO TAKERU POINTED OUT THAT ALTHOUGH THE KOJIKI AND NIHON SHOKI MENTIONED THAT THE FOUR GENERALS AND JAPANESE TAKEZUN AND OTHERS HAD MERGED EAST JAPAN INTO THE TERRITORY OF THE JAPANESE MONARCHY THROUGH MULTIPLE EXPEDITIONS SHE BELIEVED THAT IT WAS ACTUALLY KUNUNOKUNI LOCATED IN THE NOBI PLAIN THAT FORMED AN ALLIANCE IN EAST JAPAN MAINLY CONSISTING OF REAR GRAVES IN THE PAST AND THAT AFTER THE DEATH OF HIMIKO THEY LOST THE BATTLE WITH THE YAMATAI KINGDOM OR NEGOTIATED PEACE UNDER THE LEADERSHIP OF THE YAMATO KINGSHIP IN THE END THE ALLIANCE OF THE DOGSLAVE KINGDOM AND THE ALLIANCE OF THE EVIL HORSE KINGDOM MERGED INTO ONE FORMING THE JAPANESE MONARCHY THEN THE LEADERS OF EAST AND WEST JAPAN WERE DISTINGUISHED BY THE ZENPOKOEN FUN AND THE FRONT AND REAR TOMBS FROM THE END OF THE FOURTH CENTURY TO THE FIFTH CENTURY LARGE ZENPOKOEN FUN RESEMBLING KOFUN IN MOUNT TAIDABEGAN TO APPEAR IN PRESENT DAY GUNMA PREFECTURE LARGE ZENPOKOEN FUN HAVE ALSO BEEN BUILT IN PLACES SUCH AS YAMANASHI PREFECTURE IBARAKI PREFECTURE AND CHIBI PREFECTURE AFTER THE MID FOURTH CENTURY THE DAWA TOMBS ORIGINALLY CENTERED AROUND THE YAMATO KOFUN IN THE SOUTHEAST OF THE NARABASIN AND THE YUBEN KOFUN GRADUALLY SHIFTED TO THE DAISEN KOFUN IN THE NORTH THE LARGE ZENPOKOEN FUN OF DAWA WASTHE FURUICHI KOFUN CLUSTER AND MOZUTO MBSTHE CENTER OF GRAVITY OF THE GREAT KING TOMBS SHIFTED AGAIN IN THE FIFTH CENTURY TO THE ANCIENT CITY TOMB GROUP AND THE HUNDRED TONGUE DBIRD TOMB GROUP IN THE OSAKA PLAIN REGARDING THE REASONS FOR THE MULTIPLE TRANSFERS OF THESE LARGE ZENPOKOEN FUN BASED ON RESEARCH ON THE KUSHIJI AND NIHON SHUJI THE JAPANESE ACADEMIC COMMUNITY HAS PROPOSED VIEWSSUCH AS THE THEORY OF DYNASTY ALTERNATION AND THE THEORY OF QUESTRAN TRIBES CONQUERING DYNASTIES TO EXPLAIN ON THE OTHER HAND HIROSE KAZUO PROPOSED THAT THE STRUCTURES OF THE SAKIKOJISHIZUKI AND MAKINO ANCIENT TOMB GROUPS WERE SIMILAR AND FROM THE LATE FOURTH CENTURY TO THE SECOND HALF OF THE FIFTH CENTURY THE HEADS OF THE FOUR ANCIENT TOMB GROUPS JOINTLY PLAYED A ROLE IN THE YAMATO KINGSHIP AFTER DEATH THEY DEIFIED AS THE GUARDIAN DEITY OF THE YAMATO ROYAL POWER AND ALSO PROPOSED THAT THE CONSTRUCTION OF THE LARGE ZENPOKOEN FUN HAD A DETERRENT EFFECT ON VARIOUS REGIONS AT THE SAME TIME IN THE SECOND HALF OF THE FIFTH CENTURY NO LARGE SCALE ZENPOKOEN FUN WERE CONSTRUCTED IN AREAS SUCH AS KIBI PROVINCE UPPER MAOE AND HYUGA PROVINCE AND THE SCALE OF THE ZENPOKOEN FUN IN GOKISHI CHIDO GRADUALLY DECREASED ACCORDING TO THE INSCRIPTIONS ON JIJITHE IRON SWORD UNEARTHED FROM THE INAWAMOUNTAIN KOFUN AND THE LARGE SWORD UNEARTHED FROM THE ETAFUNAYAMA KOFUN EMPEROR YURYAKU AT THAT TIME REGARDED HIMSELF AS THE KING OF THE WORLD AND HIS AUTHORITY WAS REFLECTED IN THE FACT THAT ONLY HE CONTINUED TO BUILD LARGE ZENPOKOEN FUN IN THE JAPANESE ARCHIPELAGO IN THE SIXTH CENTURY THE HORIZONTAL CAVE STYLE STONE CHAMBER OF THE KINAI SYSTEM BEGAN TO BE POPULARIZED IN ZENPOKOEN FUN AT THE SAME TIME THEY BEGAN TO APPEAR IN AREAS SUCH AS SETSU PROVINCE OWARI PROVINCE UJIPROVINCE AND EVEN IN KANTO REGION WHICH WERE CONSIDERED AS BASES FOR THE ESTABLISHMENT OF THE EMPEROR KEITAI AMONG THEM UENO HAD A PARTICULARLY LARGE NUMBER OF ZENPOKOEN FUN COMPARED TO ONLY THIRTY NINE IN KOZUKE PROVINCE DURING THE SAME PERIOD UENO ALONE HAD NINETY SEVEN WHILE IN KANTO THERE WERE TWO HUNDRED SIXTEEN AS BOTH KINO AND OIZHANG DECREASED ONLY THE KANTO REGION CONTINUED TO CONSTRUCT LARGE ZENPOKOEN FUN THROUGHOUT THE SIXTH CENTURY INDICATING THAT THE YAMATO MONARCHY AT THAT TIME RELIED HEAVILY ON THE EASTERN KINGDOM IN BOTH ECONOMIC AND MILITARY ASPECTS FROM THE END OF THE SIXTH CENTURY TO THE BEGINNING OF THE SEVENTH CENTURY LARGE SQUARE AND CIRCULAR TOMBS REPLACED THE FRONT AND BECAME THE MAINSTREAM IN THE MIDDLE OF THE SEVENTH CENTURY THEY EVOLVED INTO OCTAGONAL TOMBS REGARDING THE REASON FOR THE CESSATION OF CONSTRUCTION OF ZENPOKOEN FUN AIICHIROSHIRAIISHIBELIEVES THAT IT IS RELATED TO THE REFORMS OF CROWN PRINCESHOTO KU AND SOGANO UMAKO AS WELL AS THE ESTABLISHMENT OF THE NATIONAL MANUFACTURING SYSTEM WHILE THE FINAL ZENPOKOEN FUN ARE BELIEVED TO BE AS A MOUNTAIN KOFUN AND RONGMACHICLAIMS THAT THE KOFUN WERE BUILT IN THE FIRST HALF OF THE SEVENTH CENTURY

Kunci

cipher	ABCDEFGHIJKLMNOPQRSTUVWXYZ
plain	RYIOVKJBXNDQFMPHETGSACWLZU

Referensi

Tabel referensi frekuensi monogram, bigram, dan trigram Bahasa Inggris yang digunakan pada langkah-langkah di atas mengacu pada [3, 4].

2.2 Horizon Chartreuse

Deskripsi Soal

Di lembar catatan riset berikutnya, Marisa menemukan laporan mengenai fenomena aneh yang menimpa langit malam. Seseorang telah mencuri bulan purnama yang asli dan menggantinya dengan purnama palsu yang redup. Laporan yang memuat kronologi hilangnya bagian dari langit tersebut kini tidak dapat dibaca karena seluruh teksnya dienkripsi menggunakan **Vigenère Cipher**.

Manfaatkan metode **Kasiski Examination** untuk mengidentifikasi perulangan pola karakter pada *ciphertext* dalam memperkirakan panjang kuncinya. Setelah panjang kunci diperoleh, terapkan analisis frekuensi bahasa Inggris pada tiap sub-teks untuk merekonstruksi kunci dan memulihkan seluruh isi naskah.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/1LyCID57V3ckzlvQ9Wnj-mGg_YMH0P50e/view?usp=sharing
- md5sum: 6b933809bbedb3c5c0f1fd9e2b1076cd
- sha256sum: d6ecf3775c56593f833db5c62e1447f2668b41a310562e7a9ccd6a1486d5f641

Berkas *Ciphertext*

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Langkah Penyelesaian

Artefak

Plaintext Hasil Dekripsi

Kunci

Referensi

2.3 Domain of Hopes and Dreams

Deskripsi Soal

Dalam suatu catatan simposium, Marisa menemukan rekaman mengenai keberadaan sebuah dimensi buatan yang terisolasi dari dunia luar. Batas-batas menuju wilayah rahasia ini dikunci menggunakan segel jimat yang dienkripsi dengan **Playfair Cipher** agar tidak dapat diakses secara sembarangan.

Tugas kami adalah memulihkan naskah petunjuk akses tersebut. Susun kembali matriks kunci dengan menganalisis pola kemunculan pasangan huruf (*bigram* maupun *n-gram* lainnya), distribusi karakter, serta tebakan kata-kata umum dalam **Bahasa Inggris**. Setelah matriks kunci berhasil direkonstruksi, dekripsikan seluruh dokumen rahasia tersebut.

Berkas lampiran:

- Attachment: <https://drive.google.com/file/d/10Dljf85XJXS41PSSqrfaqU0QyWTuneuX/view?usp=sharing>
- md5sum: b6f21903cbdb62ca899d6bb2694814b1
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Berkas *Ciphertext*

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TVBCPESACKATCUDBAPEDSACKATCUDBELFSIDGBSTUCRUDITLNUESVBVTBFGUAPFTUKCUYSGSPSGSTCUTOZMLNBQZBCCVCULALDUBDSOL
UADIUABFTUSIFSCUFEGMYAELUNESYASFBYMNBTTSDBDIOUDMYGMWTVTUZLPRMICVCYUAAHVIOBFIDINYSUAGYTFDLUBMWIDSLHKEYACBDI
NHUAQUDMIEMQUGLQSLQPYFZIZGLYIACTVBFBEYMNVBIOICTFMSLELUIBIDBYMNTGSZELCZBQTIOLVTNUFUPAULYTTQSZOSNUFTDNBLDU
YIYVYSICSQSQTFIATAGBIZTABRIOAPFTIYDIUEBPOUBELDIMSNGIUEBPOUBGINWAMGSYAQNLSPAFFIDBYMNTGUNUBBZTFFMNEQBETDYDI
HBCVGYUNOSVBFWMZVITBWUNTUBSUOCBRTNWUBBZTFMGYAQNTFMQYMTNSNGIUIZBYSZTYUADVBTFIAGLOLIPULIFIOGLOLIPDNLCDSPBG
TVSLFIIATDBUTYFTINMYAQNNIACBDPYBQYTOIATCAEBQIOPMMZCNMZTVTIINYSIOZUIFIOTFTIBFIAFPDSMSNGIUYSMSIVDUPSCXOSPR
YTPRYTTCUGRUIBGBAIPSCXMMWQIOQINWUBALICFLCXLDUMQIZSSISABANFBESYUOELDIPRYSYBPABVQMIYBELCZBQTIISHZOSNUFTDNBUB
ACITYINPOSYTATPASDQLYGBFTVUFFBDSQAYSVBWNBWMBGPAHYBRCLUIBBFBEYMNBEGBVTCBDNIYDIYVSDUYIVTGLICUAHYAYTTVOS
SFBTFIDBFTVTYYSYBBFUAFUCKVTAHBBICVUTCZQAXLBFTVDUACYSVBYIVWUBTTFMWCUTFEIXIBTYACRDSBYSYBBFUAFEQLFCVBYBT
PSVTQDUKTUYAIFQLLXQLBRBVTVCVCYUDEUAHVBEQMEVFBVBMFYAABQAYAWTTYCRDSFLBTBTFYSLRIOBFIDQDUKTUYAIFRYTYUAFEFLUKBF
CUMWUNTUQAICBFUAHPSUICLUFTPAZLMRZSUKTSGVVTCDIHBUNDGCTEGBOSVBSWSDUAUUKMSNEYAABICGCCYBYTINBTLAQSSFLAIDBUSG
SZYAXILOBUTYPRYSNWNWUBBOPAHXQMBFYAHKCRDSGCCYYBMFIESQKRQSVTYGSFGBVTRUZAMRFGBOIUAESYADNIKUFFBNIEBSIVBIDYSNI
IFNYTISLQPPRIFSZYUNBSMYBURFSYUBCYRLBFIOYCNWNUQLCONICPUCTEMZNGEONIGBWSTFIATBAUSMBFBVSZABIOATYTSBTELFTLZ
PQDYCKTUZLOIZLVTBFIDLRIOAPAMUKTVUABFDYZTAUIMVSAZBTFLZIFGBICQLFTDNYBDSUBIZAUZLMFTVTUUKCUYSGPSASLIGBSXPDTY
YSKTIZTGLNUMIFAESUUKBFTIZMWIQDACDSQDMVUNBHCYIZFTENNIWISUIKSZOLFTYGTIFTNWBWUZCUFRUFIVBTYACRBFVDUYSALWTVBE
QMEVFBGBVTIOPAMSIVBGTVTUPRITPOUBUABFTYLOBVUOBFIDMSNGIUSZQYTONUGCYBCUEAMKTVDRFTVBFTVTRNIFCUBDTIIVTAPLADBT
ICGBYAFIATBVSZABMSEYDUIMPDDUYTTEYDRNIFURBDDUYTTEYDRNIFELDIVTICLFTVTUUFFBAPXBSPPSIPCUYITAEACKSPSPSIPCUYTUASD
GBVXSZTOVTINTFOUACQIIICIOSEGBYAFIATBVSZABIAEQCUYSGPADVTYTONZIEBQDAMUKBFIDIOYBNUUCGBCKBEFLUKIFQLBFBZBTINQYOC
IZBTCBCUQSVFDEBAPWFUZBGDQBTUVTNRNIFCUBDTIIVTAPLADBTICGBYAFIATBVSZABDNLCGYAACUAVGIPPQMSYSBFDYBIAITZLPTSPMN
BEBUSGSZQYCREBIPSAYGBKIAVGTQZSOLGLOLICADQYBANFCVCTIEBCKBEGBICQLFIDPVZVTDEVBNWNIIRIZIONFAPVMAVBTINVTQSV
GVIZTBQDOIQYTAIFLICPZMQUGITQSVFSDSYBICASICFLMTIEATPACISZOLNUNUMYUAVGIPPQMSGLFLNUYGUAIOONDBNTVTIDBTMYT
IHNURUUKYSYBMWTVBVBGTAPDMEODNLCSDBGSGBSLIFGBTYTCKDYHRQLDIGBSTFTIAPRBQLEDQLYBICASAUDIAPBFTVGBWSTFNFAPZL
IMMAURYUALLQIZTESZDUBKMNNDBGMZEODQSYSPBEFLANFBELQDIUPZLSUFITPOUBTRMSIYEIEOCUYAHKSQBFBGTVYBFTDNSFMRYAABMS
LNNFBEMDBDSPBYOETITNMNGEWBYYBBFUANQNDMIETFMSNEYAABCBPBYLSPSSFINPZMQMQSXYBUSEARUAKUMZIZVBVTPACRBUAFNNWIB
NQBEQLBISUUKQAXLMRTFBKAPZLACUNIBFWMZUACVDBNWUBICNLKXPAINUMYASRSZEIUCIRSZBRSPABFIFGBAINWAMMLDUMRBUUMYAIZDU
MSNGIUNUGCGLOLAIRFTVINUMYADBSFIUANOUVTOSYNNIWIILHKNZIAMVUOUFEGPRIFQLBRZIOYNYBFTYBURTICZSEADEASIPYSMWIDLR
CRDSIOGPIINAMZTIMWDQEOZLZACRDSQAUAVGIPPQMSRSPAPZMAUBFUAFNBANFBEMAURODUNVTRSCVBUTREVTQMIECBUAQETIEIWTBTAR
ODFISDZIIIPDBACUAHVSZABUMOLCREBSIDSNUYGIZTGUABTNUGTUALZSZABRLBFAZPMVZSTTYSZIASEFLUKIDINIBQLFIPMATDUUMYASLHK
BFUZBCCVCUBFUAFNUKMEYBYIAUQBTDCBDPYBLSPSSFINIAQDVGUBPRMTMYEIEOUAHKTVSLCREBBKAUGATYEIEIKSZEIAIEBUMLSPSSF
INITUKCUYSGPMQUZDUAIEIZIUBUCPMMSLYDUIPFLAUQUSLUGIFGBMVUNNMTVASTADNNWUMDPYBDSFNWUMYALTBUUTYGBFSYBCIDBITYRL
NWBHDUQSWSSQCUEBSFIULPCUDIAUDBUTIIIVBUMYFSEVHUTUCKATCUDINWUMIPLSDUAEBVTIFBTUALSPSSFINTFMVBUIMBDBGPAHYFTIY
BTDSCBELDIRFTVQLRUUKCBCUDSNEGBTCMPOTITNMNTYGBIZFIAPRLPAUMFSIUXTFLALIBVITIFTTSPNSTDIACLRITFBSYUZBEQMYTMS
NGIUMWADVTAUBFGEACTYTVYGCOSYTFASACRBFBCSDSPSGHCYUYSYSTNBEMYNEQLIFQMYTMAQAPASDRMLFDSRTHKDIACMWIDNWBWNO
YCAIRFTVOUOYIGBFBYLRUNNWUBDQBFYDSIAGYCUFLHKTIVIFUQTYAAACYSYLAPIYDSYBULIFLSUQUADUZMWICZQASLAIEBTPVMSNGIUDM
BQTIIDERNZIAEOYAQNLQRMQLYGIDDIUABTCUNUHKEBLPCUDIIRSZYBLRIACRDSXTDUEBLZSZABMWTVBEGVTDNLCDUSNNWNIZSRINYOSSF
QSWTSKWLGLOLIPIAEOYAQNTFQLFIMWIDQITFFLIRIUBFTIUCDBPROIVSDSNUYGYOTSTFINUMSTBKNUYGIRVTDYHRSZSTTYBFIDYLBKTI
SFIUAPIVYBBFAQOIVTTAKRGVTICELDSIVSLACOIVTINNFTYADVMTSLGMAVSMAUBSIMNIAAUAPLPNQUAUGSIBUYGISNTFLICGBSXPDTYYS
GHURUAIYYOTSTFLINWUMSTBKNUYGYMTYELBRQLDSRTCBFTIDPLAFSSUIRLNAMHSTILATBSIDIACUAFYLDYAFSBUUFASLXTFLIAMODYMN
BVQMSOOSYTVTNUFUPAPADYRFGVBTIOPAIRSZBFIDTICBDNIYTVBUUFASIPBFDISLYITQITFLVYOVTDQUAQUIMSAFIALAQSSFLAIDBY

```

IDEBZSSTFUAINWUMVTNUFUPAZLIFZIUUAHVTFBGBQYINSQYOTSTFLIUHZZSRBFLIYBINFTYOSSFYONIUGSIIYBEGBNIEBIFBFDYSTTYS
FSIUCGBTSDMTATYDSYCSRBDFTPADSCBDPYBYOTSTFLINWUMSTBKIAQCVBUQLNPSNUCPRYMYAUCBTDSIYBEGBSZUTOIACPSZIIIMWNTFL
MRQLFSBUUFASIPVBVYBIACRDSCBTFSYIVEAESTIDSOLGCIYVTBFSYTUUKCUYSGPMDDBDSPTYNEBPRIRLNUGCISDYOTSTFLINWUMLSOLEI
VTNUGCBFSYTUICGBYAFIATBSZABTFFLCXUFEBTFFLBDUSTICQLFRZSPMYAIAMVUOITCKGULDZSEVFBTFNYUBFIKIAVGBFBESONTDSBF
DYIEITTBZMWITOGBOIMYOLAIVGELFIGBNDEIINUMTIUQAQVTITTAUKUABFTYFWUABGTVYTIHIDBYMNBVTFWLSZGUYAQNLSAPFPBFIAGLOL
APUCIFGBYSSFBFIDPLNQAANUGRNIFUAVGIPPQMSLDYAESDUSIQLYGONPRYTTOUADBMFBDLOBEYQCUCYSGPSAICVIBFGYTVSEFLUKTIRTUC
IOAMUKIOBFCIZSTTVYOMZTVVTOLUAIFFLITOIYSMKTIPSRFCUIYBTEMSNGIUYSBFIDBFBGLYIAPTYAABAPECTVLNBZTVIAEAYTQDPT
PAYGAUCUBIMYSYIAUDDNAIIMMDBDINUBQDQYATYBCSEGBZLSEQMEVFLFIAPWSINDSDPLACRBFYIOBFVTGVAQITITEBQSQBFVTCYDYMN
BEQLFIPAUMQSVXGBLKYIAPTYAICQYTOYIZLQPPSSTOUIQZSKVQYQDAMDNFTPAUABFDYIEITTBZMWITAGBIZTABFIOIDUAIKVUADUFIBF
TVNTTYGADUXNSYTFURBEFLCXUAQYIYONTASIYUMOLAUNNYFTYGLZMSRFLCXUAHEMYBUNUUKBFUAFYFLUKUAQYICGBNUUCQAICMSNGAP
RLOBSAUKBFTIQSVSYA

```

Langkah Penyelesaian

Playfair Cipher sangat berkaitan dengan penggunaan bigram (karena aturan enkripsinya berpusat pada pemetaan bigram di sebuah matriks kunci 5×5). Kelemahan *cipher* ini:

- Rentan terhadap analisis frekuensi **bigram**, jika teks *cipher* cukup panjang.
- Bigram dan kebalikannya akan menghasilkan *cipher* yang saling berkaitan: jika bigram plaintext XY (baik hubungan searah baris, kolom, maupun *rectangle*) dienkripsi menjadi *cipher* PQ , maka bigram kebalikannya YX akan dienkripsi menjadi QP , kebalikan dari *cipher* sebelumnya.

Penyelesaian soal ini dibagi menjadi dua tahap. Tahap pertama adalah **analisis frekuensi bigram secara manual** untuk menambatkan sebagian huruf pada matriks kunci. Tahap kedua adalah **pencarian otomatis dengan *Simulated Annealing* (SA)** untuk melengkapi sel-sel yang tersisa, karena melanjutkan penebakan manual sel demi sel terlalu lambat dan mudah tersesat.

Tahap 1: Analisis Frekuensi Bigram

1. Hitung frekuensi bigram **non-overlapping** pada *ciphertext* (`Solver.freq_bigram` pada `src/23/solver.py`). Perhitungan ini dilakukan pada *ciphertext* **sebelum** dilakukan substitusi apa pun, sehingga hasilnya tidak berubah walau hipotesis matriks kunci diubah-ubah. Terdapat **392** bigram unik, dengan sepuluh besar sebagai berikut.

Bigram (non-overlapping)				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	BF	3.24	TH	3.56
2	UA	1.95	HE	3.07
3	GB	1.87	IN	2.43
4	VT	1.62	ER	2.05
5	TV	1.60	AN	1.99
6	YA	1.45	RE	1.85
7	IA	1.34	ON	1.76
8	DS	1.22	AT	1.49
9	ID	1.22	EN	1.45
10	QL	1.21	ND	1.35

2. Susun hipotesis awal dari empat pencocokan berikut:

Cipher	Freq (%)	Dugaan <i>Plain</i>	Dasar Dugaan
BF	3.24	TH	bigram tersering pada kedua sisi
GB	1.87	HE	peringkat 3 cipher vs bigram Inggris tersering kedua
VT	1.62	RE	pasangan resiprokal dengan TV
TV	1.60	ER	pasangan resiprokal dengan VT

UA belum dicocokkan karena cukup sulit memetakan huruf *plain* yang beririsan. Melanjutkan ke VT/TV, keduanya muncul dengan frekuensi yang mirip, sehingga hipotesis selanjutnya adalah memetakan VT/TV

ke **RE/ER**, karena memang keduanya merupakan pasangan bolak-balik yang sama-sama sering muncul dalam Bahasa Inggris (2.05% dan 1.85%). Pencocokan **GB** → **HE** juga terverifikasi lewat kemunculan **BG** → **EH**.

3. Turunkan hubungan posisi antarhuruf dari keempat hipotesis tersebut. Ketiganya ternyata merupakan hubungan *rectangle*, sehingga masing-masing mengunci baris dan kolom relatif antarhuruf:

- **BF** → **TH**: B sebaris dengan T, dan F sebaris dengan H.
- **GB** → **HE**: G sebaris dengan H, dan B sebaris dengan E.
- **TV** → **ER**: T sebaris dengan E, dan V sebaris dengan R.

Gabungan ketiganya membentuk blok T-E-B pada satu baris, F-G-H pada baris lain, serta R-V pada baris ketiga, dengan kolom yang saling bersesuaian. Penambahan **TF** → **IT** kemudian menempatkan I. Total **9 dari 25** sel berhasil ditambatkan secara manual, dan inilah isi **GRID** pada `src/23/solver.py`.

Catatan: hipotesis sempat keliru ketika R dan V diletakkan pada baris yang sama dengan T, E, dan B. Susunan itu memaksa **BE** (0.88%) terbaca sebagai **VB**, sebuah bigram yang praktis tidak muncul dalam Bahasa Inggris. Kontradiksi inilah yang menjadi dasar penolakan hipotesis tersebut.

Hasil tahap ini adalah hipotesis matriks kunci berikut, dengan sel kosong menandakan posisi yang belum dapat disimpulkan dari analisis bigram:

I				
T	E	B		
F	G	H		
R	V			

Tahap 2: *Simulated Annealing*

Melanjutkan tahap 1 secara manual untuk 16 sel sisanya tidak efisien. Seringkali kesalahan posisi satu huruf baru ditemukan beberapa langkah kemudian. Karena itu, sisa pencarian diserahkan pada *Simulated Annealing*.

1. Hasil dekripsi dinilai dengan **fungsi objektif** berupa **log-probabilitas quadgram** (`Solver.score`), yaitu

$$S = \sum_i \log_{10} P(q_i),$$

dengan q_i adalah quadgram ke- i (jendela 4 huruf yang saling tumpang-tindih) pada hasil dekripsi. Nilai P diambil dari tabel **QUADGRAM** pada `src/frequencies.py` (*quadgram* yang tidak terdaftar diberi nilai dasar 0.001 sebagai penalti). *Quadgram* dipilih karena menilai kelayakan rangkaian huruf yang lebih panjang daripada bigram, sehingga lebih tegas membedakan teks Inggris dari teks acak.

2. Satu langkah dilakukan dengan menukar isi dua sel bebas secara acak (`Solver.swap`). Penukaran antarbaris atau antarkolom penuh hanya diaktifkan bila seluruh sel bebas, karena operasi tersebut akan menggeser sel-sel yang sudah dikunci.
3. Solusi tetangga dengan skor lebih tinggi selalu diterima. Kemudian, solusi yang lebih buruk tetap diterima dengan peluang $\exp(\Delta S/T)$, sehingga pencarian dapat keluar dari optimum lokal. Suhu awal $T = 20$ diturunkan bertahap sebesar 0.5 hingga mencapai 0, dengan 2500 percobaan pada tiap tingkat suhu (total ± 100.000 evaluasi per *restart*). Proses ini diulang sebanyak 2 *restart*, dan skor terbaik yang akan dipakai.
4. Selama pencarian, penilaian hanya dilakukan pada **2400 karakter pertama** *ciphertext*. Ukuran ini sudah cukup untuk membedakan kunci benar dari kunci salah, namun memangkas waktu secara signifikan. Dekripsi penuh baru dijalankan sekali di akhir memakai kunci terbaik.

Hasil

Pencarian konvergen ke skor -6877.2 dengan waktu eksekusi sekitar **2 menit 30 detik**, dan menghasilkan pemetaan lengkap untuk seluruh **392** bigram unik. Pengujian berulang memberi kunci yang sama secara konsisten.

Perlu dicatat bahwa penguncian sembilan sel hasil tahap 1 sangat menentukan. Ketika seluruh 25 sel dibiarkan bebas, pencarian dengan jumlah *restart* yang sama kerap berhenti pada optimum lokal (misalnya pada skor -7098.3) dengan hasil dekripsi yang tidak terbaca. Dengan kata lain, analisis manual dan pencarian otomatis saling melengkapi: analisis manual memperkecil ruang pencarian dari 25 sel menjadi 16 sel, sementara SA menyelesaikan bagian sisanya.

Artefak

Skrip *solver* yang digunakan untuk menghitung frekuensi bigram *ciphertext*, menjalankan *Simulated Annealing* untuk melengkapi matriks kunci, menurunkan pemetaan bigram *cipher* $\rightarrow$ *plain* berdasarkan aturan Playfair, mendekripsi *ciphertext*, serta mencatat setiap hasil dekripsi ke `log/plain/` dan metadatanya (pemetaan, matriks kunci, frekuensi bigram, dan skor akhir) ke `log/metadata/`.

```

1 import json
2 import math
3 import os
4 import random
5 import sys
6 from datetime import datetime
7 from typing import Dict, List, Tuple
8
9 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
10 from frequencies import QUADGRAM
11
12 ALPHABET = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
13
14 _ = 0
15 GRID = [
16     [_, _, _, _, _],
17     ["I", _, _, _, _],
18     ["T", "E", "B", _, _],
19     ["F", "G", "H", _, _],
20     ["R", "V", _, _, _]
21 ]
22
23 class Solver:
24     def freq_bigram(self, string: str) -> Dict:
25         freq_data = {}
26         total_ngrams = len(string) // 2
27         for i in range(0, len(string), 2):
28             ngram = string[i:i+2]
29             freq_data[ngram] = freq_data.get(ngram, 0) + 1
30         for ngram in freq_data:
31             freq_data[ngram] = (freq_data[ngram] / total_ngrams * 100)
32         return dict(sorted(freq_data.items(), key=lambda item: item[1], reverse=
33             True))
34
35     def find_pos(self, grid: List, letter: str) -> Tuple[int, int]:
36         for r, row in enumerate(grid):
37             for c, val in enumerate(row):
38                 if val == letter:
39                     return r, c
40         return None

```

```

40
41 def decrypt_pair(self, grid: List, a: str, b: str) -> str:
42     ra, ca = self.find_pos(grid, a)
43     rb, cb = self.find_pos(grid, b)
44     if ra == rb:
45         return grid[ra][(ca - 1) % 5] + grid[rb][(cb - 1) % 5]
46     if ca == cb:
47         return grid[(ra - 1) % 5][ca] + grid[(rb - 1) % 5][cb]
48     return grid[ra][cb] + grid[rb][ca]
49
50 def build_mapping(self, grid: List, bigrams: List) -> Dict:
51     return {bigram: self.decrypt_pair(grid, bigram[0], bigram[1]) for bigram in
    bigrams}
52
53 def decrypt(self, string: str, mapping: Dict) -> str:
54     return "".join(mapping[string[i:i+2]] for i in range(0, len(string), 2))
55
56 def score(self, plain: str) -> float:
57     total = 0.0
58     for i in range(len(plain) - 3):
59         total += math.log10(QUADGRAM.get(plain[i:i+4], 0.001))
60     return total
61
62 def random_grid(self, grid: List) -> List:
63     used = [val for row in grid for val in row if val != 0]
64     rest = [letter for letter in ALPHABET if letter not in used]
65     random.shuffle(rest)
66     return [[val if val != 0 else rest.pop() for val in row] for row in grid]
67
68 def free_cells(self, grid: List) -> List:
69     return [(r, c) for r in range(5) for c in range(5) if grid[r][c] == 0]
70
71 def swap(self, grid: List, free: List) -> List:
72     new_grid = [row[:] for row in grid]
73     if len(free) == 25 and random.random() < 0.1:
74         a, b = random.sample(range(5), 2)
75         if random.random() < 0.5:
76             new_grid[a], new_grid[b] = new_grid[b], new_grid[a]
77         else:
78             for row in new_grid:
79                 row[a], row[b] = row[b], row[a]
80     return new_grid
81     (ra, ca), (rb, cb) = random.sample(free, 2)
82     new_grid[ra][ca], new_grid[rb][cb] = new_grid[rb][cb], new_grid[ra][ca]
83     return new_grid
84
85 def anneal(self, string: str, bigrams: List, grid: List, temp: float = 20.0,
86             step: float = 0.5, rounds: int = 2500) -> Tuple[List, float]:
87     free = self.free_cells(grid)
88     current = self.random_grid(grid)
89     current_score = self.score(self.decrypt(string, self.build_mapping(current,
    bigrams)))
90     best, best_score = current, current_score
91     while temp > 0:
92         for _ in range(rounds):
93             trial = self.swap(current, free)
94             trial_score = self.score(self.decrypt(string, self.build_mapping(
    trial, bigrams)))
95             delta = trial_score - current_score

```

```

96         if delta > 0 or random.random() < math.exp(delta / temp):
97             current, current_score = trial, trial_score
98             if current_score > best_score:
99                 best, best_score = current, current_score
100         temp -= step
101         return best, best_score
102
103     def solve(self, string: str, grid: List, restarts: int = 2) -> Tuple[List,
104     float]:
105         sample = string[:2400]
106         bigrams = list(self.freq_bigram(sample))
107         best, best_score = None, -math.inf
108         for _ in range(restarts):
109             result, result_score = self.anneal(sample, bigrams, grid)
110             if result_score > best_score:
111                 best, best_score = result, result_score
112         return best, best_score
113
114     def load_ciphertext(base_dir: str) -> str:
115         with open(os.path.join(base_dir, "f.txt")) as f:
116             return f.read().strip()
117
118     def write_plain(base_dir: str, timestamp: str, decrypted: str) -> str:
119         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
120         os.makedirs(os.path.dirname(path), exist_ok=True)
121         with open(path, "w") as f:
122             f.write(decrypted)
123         return path
124
125     def write_metadata(base_dir: str, timestamp: str, mapping: Dict, grid_manual: List,
126     grid_filled: List, freq: Dict, score: float) -> str:
127         path = os.path.join(base_dir, "log", "metadata", timestamp + ".json")
128         os.makedirs(os.path.dirname(path), exist_ok=True)
129         payload = {
130             "timestamp": datetime.now().isoformat(timespec="seconds"),
131             "mapping": mapping,
132             "grid_manual": grid_manual,
133             "grid_filled": grid_filled,
134             "freq_ngram": freq,
135             "score": score,
136         }
137         with open(path, "w") as f:
138             json.dump(payload, f, indent=2)
139         return path
140
141     if __name__ == "__main__":
142         base_dir = os.path.dirname(os.path.abspath(__file__))
143         content = load_ciphertext(base_dir)
144
145         solver = Solver()
146         freq = solver.freq_bigram(content)
147         grid, score = solver.solve(content, GRID)
148
149         mapping = solver.build_mapping(grid, freq)
150         decrypted = solver.decrypt(content, mapping)
151
152         timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
153         plain_path = write_plain(base_dir, timestamp, decrypted)
154         metadata_path = write_metadata(base_dir, timestamp, mapping, GRID, grid, freq,

```

```

154     score)
155     print(decrypted)

```

src/23/solver.py

Plaintext Hasil Dekripsi

PARTIWHATCHANGEWILLTHENNEWPOWERSBRINGTOGENSOKYOIDLIKETOTHANKEVERYONEFORTAKINGTHETIMEOUTOFTHEIRBUSYLIVESTOPARTICIPATEIHIEDANOAKYUXUWISHTOCOLLECTYOURVALUEDVIEWSANDREPORTTHEMFORTHEBENEFITOFALXLGENSOKYOWITHTHATINMINDIDLIKETOTURNITOVERTOYOURSYMPOSIUMSMODERATORHUMANMAGICIANANDONEOFGENSOKYOSLEADINGEXPERTSONYOUKAIMARISAKIRISAMEAHUMIAMTHEMODERATORMARISACAUSEUHIEMANBECAUSEIBELIEVEDTHESYMPOSIUMTOBEPRESENTEDHEREMAYBEINTERESTINGIHAVEACXCEPTEDTHEHONOROFTHISROLEANDUMXIUSTSPEAKASYOUNORMALXYDOHEARINGYOUOTALKLIKETHISISDISGUSTINGOKAYTHENLETSTARTWITHTHEINTRODUCTIONSTAKEITAWAYIAMKANAKOYASAKAIMANAGETHESHINEONYOUKAIMOUNTAINUSUALLYTHINKOFNOTHINGBUTSHINTOBUTIAMINTERESTEDTOHEARFROMTHOSEOFOTHERXRELIGIONSTODAYIAMBYAKURENHIXIXIRIXIAMTHECHIEFPRIESTOFTHEMYOURNTEMPLEONTHEOUTSKIRTSOFTHEVILXLAGEALTHOUGHIMTHECHIEFPRIESTIMSTILLANOVICEYETXTOATTAINENLIGHTENMENTBUTPLEASECONSIDERMYVIEWSTODAYIAMTOYOSATOMIMINOMIKOUMPRESENTLYIVERENOUNCEDMYHUMANITYANDAMAHERMITIONLYAWOKERECENTLYSOIMSTILLHALFASLEXEPBUTIFELTTTHATIMIGHTDEEPENMYUNDERSTANDINGOFGENSOKYOBYATXTENDINGTODAYSOWASDETERMINEDTODOSOSPEAKINGOFWHICHTHERESIDENTSOFGENSOKYOSEXEMTOBEONGOODTERMSIFTHEYCANHAVEASYMPOSIUMLIKETHISWHOATHISISIUSTTHEFIRSTONEBUTWEREALWAYSUPPERTAKINONNEWCHALXLENGESTHUGHISTHATSOWHATCHANGEWILXLTHENNEWPOWERSBRINGTOGENSOKYOSOWEGOTHEREASHINTOGODABUDXDHISTPRIESTANDATAOISTHERMITSINCEWEVEPRETTYMUCHGATHEREDUPABUNCHOFBUSINESSSRIVALSTHISUGHTABEGOXODATHREXWAYTALKBETWEXENTHREERELIGIOUSFIGURESXSOWHATAREYALLGOINGTOTALKABOUTARENYOOUOURMODERATORSUPPOSEDTODECIDETHATOHYEAHOKAYTHENTHEFIRSTXTHEMEISWHATCHANGEWILLTHENNEWPOWERSBRINGTOGENSOKYOIUSTTHROWXWHATEVERYAWANTOUTXTHEREASYOUALREADYARELAUGHTHENNEWPOWERSREFERSTOALXLOFUSRIGHTAHRIGHTIUSTXTOCLARIFYALITXTLEYALLTHREXARETHENNEWCOMERSTHEHIDDENMEANINGOFTHESYMPOSIUMISTHATYAMIGHTBEHOSTILEELEMENTSHERETODISRUPTOURPEACEFULGENSOKYOORSOMETHINANDHEREITHOUGHTYOUWERESTARTINGTOGETUSEDTOMEDONTMATTERHOWMANYEARSGOBYNEWCOMERSWILXLALWAYSNEWCOMERSFROMTHEOLDTIMERSPERSPECTIVEANYWAYWELLTHENASTHEREALNEWCOMERISHALXLSTARTGOFORITMYFIRSTIMPRESSIONOFGENSOKYOBESIDESTHEFACTTHATTHEAREYOUKAIWHOCAUSEHARMTOMANKINDISTHATXTHEMINDSOFHUMANSARECAPRICIOUSANDEASILYSWAYEDTHEHUMANSOFTHEVILXLAGEPENDTHEIRDAYSINMENTALVAGRANCYFALLINGTOIMPROVETHEIRLOTINLIFEANDTHEYOUKAIAREALXLOWEDTOTAKEADVANTAGEOFTHEIRCAPRICIOUSNESSTOTHEIRHEARTSCONTENTFRANKLYITSAMIRACLETHATSOCIETYHASNTCOLLAPSEDEYETANDIBELIEVETHECURRENTBALANCEISDANGEROUSLYUNSTABLEUHGUESXSXWHYDOESGENSOKYOHAVENOADMINISTRATORSOCIETYCANXNOTFUNCTIONWITHALXLHUMANSPOSSESXSINGTHESAMESOCIALSTANDINGANDRANKANADMINISTRATORHATSRIGHTHUMANSNEXEDLEADERSTOGUIDETHEMTOWARDSAHOPEFILXLEDFUTUREIFNOXONEXELSEISQUALIFIEDTHENIWOULD CERTAINLYIWONDERWHATKINDOFFXUTURELIESINSTOREFORTHEHUMANSUNDERTHERULEOFSOMEONEWHOISNOLONGERHUMANRULEISSUCHANUGLYWORDITWOULDJUSTBEASTOPGAPMEASUREUNTILOMEONEFROMAMONGTHEHUMANSISABLETOIMITATEANDOVERTURNMEMAYISAYAWORDYOU SAYTHAT THEREISNOADMINISTRATORINGENSOKYOBUTIDONTTHINKTHATSSQUITETRUWHATDOYAMEANGENSOKYOSCURXRENTFORMCANXNOTBEMAINAINEDWITHOUTTHEYOUKAIISPOWERGENSOKYOWASCREATEDBYXYOUKAITOSAVETHEMSELVESAFTERALXLTHEHUMANSINGENSOKYOEXISTFORNOFURTHERPURPOSETHANTOPRESERVETHEYOUKAIITHEHUMANSAREBEINGRULEDBYTHEYOUKAI IAGREXETHEYOUKAIKEEPWATCHONEACHOTHERTOPREVENTOTHERYOUKAIFROMRUNNINGRAMPANTANDRULINGOVERTHEHUMANSANDTOMAKESURETHEHUMANS DONTAWAKENTOTHEIRACTUALCONDITIONANDTURNDOWNTHEWRONGPATHTHEREFORETHEFACTTHATARULERHASNTEMERGEDFROMAMONGTHEHUMANSISINGENSOKYOSBESTINTERESTSOTHATSHOWITISBUTIFTHATSTHECASETHENTHEPOXORHUMANSHEREAREPITIABLECREATURESTHINKOFITXTHISWAYARETHEANIMALSINAZOXOUNFORTUNATEANDAZOOISAINSTITUTIONWHEREENDANGEREDANIMALSAREPROTECTEDANDHUMANS CANOBSERVE THEMANDLEARNABOUTXTHEMTAKINGANATURALLYFREXEBIRDANDLOCKINGITUPINSMALXLCAGEWHENYOULOOKATITFROMTHATPERSPECTIVEITSEXEMSQUTEUNFORTUNATEBUTNOWITDOESNTHAVETOWORRYABOUTPREDATORSORRUNNINGOUTOFFXOODANDTHEREWILXLALWAYSBEPLENTYOFHUMANSFORITTOLOOKATSOITWILXLENEVERGETBOREDIFYOUTHINKOFITLIKETHATARENTXTHEYFORTUNATESOGENSOKYOSHUMANSAREANIMALSINAZOXOWHATHAPPENEDTOHUMANIDIGNITYDIGNITYHUHIDONTTHINKTHATSSOMETHINGYOURECEIVEFROMOTHERSTHOUGHWHATDOYOUTHINKHMXMXMEWHATSDIGNITYLIKEPRIDEOORSELFRESPECTOFCOURSEIMPROUDOFEVERYTHINGIDOBUT EVENIF IWEREFORCEDINTASOMETHINIDBEPERFECTLYFINEWITHTHATSOLOGASIWEREHAVINFUNISEXETHENPLEASEFORGETABOUTWHATISAIDWAITWHATWEREYATALKINABOUTLAUGHTHEREASONKANAKOCAMETOGENSOKYOANYWAYIWASHAVINAHARDTIMEFOLXLOWINTHATHONESTLYIDONTGIVEAFLIPABOUTTHEHUMANSOFTHEVILXLAGE TALKABOUTYERSELVESTHETHEMEWASWHATAREYOUBRINGINGTOGENSOKYOAFTERALLSOFIRSTOFALLWHYDYAMOVETOGENSOKYOANYWAYICAMETOGENSOKYOTOSEEKNEWFAITHINTHEOUTSIDEWORLDSDHRINESHAVEBECOMEMERETOUISTXTRAPSORCALLEDNONSENSICALTHINGS LIKEPOWERSPOTSSOPLACESFORHUMANSTOXOFFERTHEIRFAITHTOTHEGODSHAVEDISAPXPEAREDGODSAREBEINGSTHATWILLCEASETOEXISTWITHOUTHUMANSBELIEVINGINTHEMSOYASAYBUTIDONTTHINKYAVEGOTTENMUCHFAITHINGENSOKYOETHEIRIUSTBELIEVINGTHATWEREREALISENOUGHINTHEOUTS

IDEWORLDTHEYVESTOPPEDBELIEVINGTHATGODSEVENEXISTINDEXEDEVERYTHINGREALLYCHANGEDONCEWECAMETOGENSOKYOHHEREHUMAN
SBELIEVETHATGODSEXISTASANEVERYDAYFACTWELXLEVENIFIBELIEVEDINYAORNOTYERRIGHTHEREAINTYATHATSNOTGOXODENOUGHFO
RTHEOUTSIDEWORLDDEVENIFYOUPERFORMMIRACLERIGHTBEFORETHEIREYESTHEYLXLCONVINCETHEMSELVESITSANILLUSIONANDABSOL
UTELYREFUSETOACXCEPTTHATITACTUALLYHAPXPENEDHUMANSAREEASILYSWAYEDCREATURESXSOTHATSTHELATESTFADAFADYESANENTI
RELYINTENTIONALFADBUTYOUKNOWTHEBESTPARTISTHATEVENWHILETHEYDONTTHINKTHATGODSORYOUKAIEXISTTHEYSTILLBELIEVEIN
ALIENSOHNEVERHEARABOUTXTHATTHEMOMENTXTHATHUMANSREALIZEDTHATTHEVASTNESSOFSPACEWASBEYONDTHEIRABILITYTOCONTR
OLTHEYSTARTEDTOCONVINCETHEMSELVESTHATXTHEREMUSTBEALIENSXSOMEWHEREOUTTHEREXEVENALTHOUGHNOBODYSCONFIRMEDITYETR
ATIONALISMSUDXDENLYTURNSTOMYSTICISMWHENITGOESTOXOFARFUNXNYISNTITIHAVEBEXENCURIUSABOUTTHISFORAWHILENOWBUTY
OURSHRINEHASTWOGODSAMIRIGHTWHYISTHATTECHNICALXLYWEVEGOTXTHREEBUTNEVERMINDTHATFORNOWOURSHRINESBEXENTHROUGHQ
UITEABITWELXLIUSTLETMESAYSUWAKOISAGOODPARTNEROOHIKNOWTHISONESUWAKOSTHEONEWHOREALLYRULESTHESHINERIGHTANDYE
RIUSTXTHEONERUNXNINSTUFXFOUTINFRONTAHEMALTHOUGHTHATSNOTQUITEHOWITWORKSITSNOTWRONGTHATXTHEMORIYASHRINEBELON
GSTOSUWAKOBUTSUWAKODOESNTHAVEMUCHINTERESTINGATHERINGFAITHYOUSEESOIHAVENOCHOICEBUTXTOGATHEFAITHFORUSBYMYSE
LFNOCHOICESPEAKINGOFWHICHMSKANAKOYOUAREFROMSUWAOFSHINANOAREYOU NOTIMNOTBUTSANAEANDSUWAKOAREILIVEDTHEREFORAL
ONGTIMETOXOHTHOUGHACTUALXLYIMALSOFROMSHINANO THATBRINGSBACKSOMANYMEMORIESOHEREALXLYWHENEVERICAMETOSUWATHELAKE
WASALWAYSFULLOFSUCHCLEANWATERITWASSUCHABEAUTIFULPLACEWHENWASTHISIMNOTSUREMAYBEABOUTATHOUSANDYEARSAGONOWADA
YSITSMADEANCOMPLETEABOUTFACEANDBECOMEARATHERCOLD CITYALTHOUGHIMPETTYPROUDOFITSEXCELXLENTFACTORIESANDTHEPRE
CISIONMACHINERYTHEYMAKETHEREAREALSOSOMENICEHOTSPRINGSBYTHETRAINSTATIONTHECONSTRUCTIONOFTHEMYOURENTempleWHO
ALETSNOTGETCAUGHTUPINTALKOFSOMECOUNTRYSIDETOWNNOONEHEREKNOWSABOUTNEXTUPISYOUPPOINTINGATBYAKURENWHYDOYOUOMET
OGENSOKYOIAMUSTASIMPLEMONKINTRAININGBUTBECAUSEOFAMINORDISAGREEMENTIWASBETRAYEDBYHUMANSANDSEALEDAWAYHOWEVE
RSOME OFMYOLDFRIENDSWHOMISXSEDMECAMETOMYRESCUEANDTHATPLACEIUSTHAPPENEDTOBEGENSOKYONOWWAITDOESTHATMEANYOUREH
ERE BYCOINCIDENCE NOTCOINCIDENCE BUT RATHERINEVITABILITYWERETHEYNOTINGENSOKYOMYFRIENDSWOULDHAVEBEXENUNABLETOMA
INTAIN THEIRSTRENGTHYOU BUILT YOURTEMPLE DIRECTLY ABOVE WHERE I WASXSLXEPINGWASTHAT ALSOINEVITABLEWHATMIGHTY YOU BETA
L KINGABOUTOHEYEAHIKOSGRAVE WAS UNDER THE TEMPLE WASNTITYOUBUILTATEMPLEFORBUDXDHISMARELIGIONIUSED FORPOLITICALMEA
SURES ONTOPOFITANDI HEARDIT WASNOTIUSTANYTEMPLE BUTAYOUKAI TEMPLEFULXLOFYOUKAIMONKSATEMPLE FILLED WITH SUCHDETESTA
BLEXEVILPOWERHOWRUDEWHILEITMAYBEFILXLEDWITHYOUKAI THOSEYOUKAIAREMORESINCERETHAN HUMANSIUSTASHUMANS CANBECOMEH
ERMIT SANDCELESTIALS YOUKAI CANBECOME GODS ANDBUDDHASMYOWNPOWERISALSOASVASTASITISTHANKSTOYOUKAI SOYOUAREAMONKPOX
SEXS XSED BYEVIL THENAMONKWHO HASFALXLENUNTHINKABLY FARORSOIDLIKETOSAY BUTASXSOMEONEWHOUSED BUDXDHISMFORPOLITICALE
NDSISUPXPOSEIDONOTHAVETHERIGHTTODOSOWELXLTOTOBHONESTWEBUILTTHETEMPLETHEREBECAUSEOURTREASUREHUNTINGMOUSEREPO
RTEDTHAT THERESSOMETHING INCREDIBLEBURIEDXDOWN THEREANDSOWESURVEYEDITWHENITTURNEDOUTXTOBESOMETHINGTHATSEXEMED
MORE DANGEROUS THANWHATWEWEREXEXPECTINGWETHOUGHTWESHOULDDOOURBESTXTOHIDEITSOI WASTHATDANGEROUSXSEEMINGTHNGTH
ENWHILEIDONTTHINKMYINTUITIONWASWRONGIREGRETXTABLY LACKEDTHEPOWERMEANINGYOU COULDN'TKEEPMESEALEDUNFORTUNATELYF
OROURNAMELESSXSOCALXLEDMONKWHOTHOUGHTXTOKEEPMESHUTOFFXFOREVER THEREVIVALOFTHE SAINTYOU CANFIGHTLATER SWEATSOLA
STUP ISMIKOSXSTORYIPUTMYSELF TOSLEEP WITHASPELXLIHADPLANXNEDTOWAKEUPASAHERMITANDENIOYTHEWORLDASIDESIREDBUTIAW
OKEONCEI WASAWARETHAT I WASNO LONGER IN THE ORDINARYWORLDITWASTHENTHATICAMETO LEARNI WASHERE INGENSOKYOIMIN THE PROCES
SOFCAREFULLYSTUDYINGTHISWORLDANDTODAYWILLBEPARTICULARLYINFORMATIVEISTILXLAVENTFIGUREDOUTXTHEREASONIMHERET
HOUGH THEREASONYOU'RE HEREISXSIMPLE IN THE SAMEWAYTHATYOUKAIORGODSLIKE MEENDUPHEREAFTERTHEIREXISTENCEIS RENOUNCED I
N THEOUTSIDEWORLDYOUVEALSOCEASED TOEXISTIVECEASED TOEXIST BUTAREN'T IHERERIGHTNOWCOMMONSENSEANDUNCOMMONSENSEINOT
HER WORDSREALITYANDFANTASYTHEOUTSIDEWORLDANDGENSOKYOARESEPARATEDALONGTHATSAMELINEHENCETHINGSTHATHAVEDISAPPE
ARED BEENFORGOTTENORHAVEHADTHEIREXISTENCE RENOUNCED IN THEOUTSIDEWORLD CANBEFOUND INGENSOKYOSOTHATMEANSI WAS COMPL
ETELYFORGOTTENSOIVECHANGEDFROMAHUMANTOAGODLIKEEXISTENCE THEN THATSCORRECTWELXLASFARASIKNOWYOUWEREALREADYR
EGARDEDASAGODINSHINTOXOHYESIFORGOTTOMENTIONTHISBUTIDONTNORMALXLYLIVEINGENSOKYOHUHLIVEINASPACEICREATEDMYSE
LFLIVINGSOMEWHEREWHERE NOBODYCAN BOTHERTHEMISTHE ESSENCE OFWHAT AHERMIT ISNOWTHATCHAMENTIONITANOTHERHERMITKNOWA
LSOLIVESINAHARDTO REACHPLACEHOWDYADOSOMETHINLIKETHATANYWAYITSSIMPLEIFYOUSTUDYENOUGH OFTHEHERMETICARTSWOULDYO
ULIKETOBECOME MYDISCIPLEUHEMMETHINKABOUTITYOUCOULD BECOMEIMMORTALOXOHOPROSELYTIZINGNOWAREWEDONTXTHINKILXLT
AKETHATLYINGDOWNSWITCHNOWANDILLTHROWINEXCELXLENTLUCKONALLYOURFORTUNESFORTWOMONTHSTHENALXLOWMEASWELLYOUAREA
YOUKAILIKEHUMANAFTERALLYOUCERTAINLYAREPOPULARLAUGHAXANYWAYIGUESXSTHATWRAPSUPTHESELFINTRODUCTIONSSOWHATI WAN
TXTOASKYALLNOWISWHADYAWANTXTODOINGENSOKYOFROMNOWONITHINKIDLIKETOLIVEACAREFREXELIFEWITHOUTBEINGDISTURBEDIZ
ZATSOTRASCENDINGWORLDLYMATXTERSTOLIVEAREFINEDLIFEINAWORLDTWITHNOWORXRIESORANYTHINGATALLTHATSMYIDEALYOU MIGH
THAVEBEXENTHINKINGYOUVE MANAGEDTOBECOMEIMMORTALANDAGELESXS BYSTUDYINGHERMITXTECHNIQUESBUTUNFORTUNATELYIMMORT
ALSINGENSOKYOHAVEANATURAL ENEMYYOU MEANSHINIGAMIYESTHEYWILLPERIODICALXLYCOMETOKILXLYOUIBELIEVETHOSEBATTLESWI
LXLBEA FORM OFENTERTAINMENTANICEISURELYSTRUGGLEINCIDENTALXLYIAMNOTINTERESTED INIMMORTALITYWELLTOBEHONESTIUS
EDTOBEBUTAFTERHAVINGBEENTOUCHEDBYMYDISCIPLESHEARTSNOWIONLYTRAINWITHANALTRUISTICSPIRITANDIWISHTOSAVETHOSEYO
UKAIWHOAREIN THE UNDESERVED POSITIONATXTHE BOTTOM OFGENSOKYOSSOCIETY SUCHASTHOSEWHO DONOTWISHTOFIGHTANDWHATLXLHAP

PENAFTERYASAVETHEMIWILLXLEADTHEMTOANEVENHIGHERSTAGESOUNDSOMINOUSTOMEIUSTIMAGINEAWORLDDXDOMINATEDBYXYOUKAIBO
 DHISATTVAXANDYOUKAIWHOHAVEATTAINEDENLIGHTENMENTASFORMEIWANTXTOCREATEAWORLDDWHEREHERECANBEAPILXLAROF SUPPORT
 FORBOTHXHUMANSANDYOUKAI THATSWHATFAITHISFAITHISNECESXSARYTOLIVESPIRITALLYRICHLIVESFREEOFWORRIESTHATMUCHISCE
 RTAINOURMETHODSAREDIFXFERENTBUTOURMINDSAREALIKESOAREYOUHOLYROLLERSPLANNINONMAKINYERSELVESRICHINTHEPROCESSO
 FGIVINUSSPIRITUALXLYRICHLIVESYESBUTWHATOFITXTHISCOUNTRYISFULLOFMISUNDERSTANDINGSABOUTRELIGIONWITHMANYPEOPL
 EWHOOPPOSERELIGIONNONEOFTHEREASONSFOR THATISBECAUSERELIGIONSTHATMISUSEFAITHARESPREADINGALSOITSANUNFORTUNATEM
 ATTERORIGINALXLYRELIGIONWASTRAININGWITHTHEGOALOF SPIRITUALGROWTHIUSTLIKESPORTSPLAYERSUSETHELATESTTHEORIESTO
 TRAIIN THEIRBODIESTOPERFECTIONTOBECOMESPIRITUALXLYRICHANDFEXELHAPXPINESXSYOUNEEDTOUSETHEORIESANDMORALSWHICHI
 SWHATRELIGIONSWEREMEANTXTOBEITSOURDUTYISTOMAKESURETHOSEINGENSOKYOBECOMESUCCEXSFULYETNOTSPIRITUALXLYIMMATU
 RELIKETHOSEINTHEOUTSIDEWORLDITHINKIGETITBUTHEARIN THATFROMYOUANYWAYSINCEYABROUGHTITUPLETSCHANGETHETOPICTOTH
 EOOUTSIDEWORLDSSHESAYSMANAGEBUTSHEACTUALLYISAGODRESIDINGINTHESHINEHERSELFATEMPLEWITHMOREYOUKAIMONKSTHANHUMA
 NONESTHESORTTHATCOULDONLYEXISTINGENSOKYOABOUTYEARSHAVEPASXSEDSINCESHECAME TOGENSOKYOINTRUTHEVERYTHINGISFINE
 ASLONGASTHEBARRIERSPOWERREMAINSTHISISASOPHISMOFTENUSED BYYOUKAI SOTHATTHEHUMANS COULDNOTBERULEDOVERANDUSEDASC
 ONVENIENTLYSOSANAECANALSO BECALLEDAGODISEEBYTHEWAYTHOUGHITSNORMALTOCOUNTXTHEMASTHREEGODSINSTEADOFTHREEPEOPL
 ETHATSONLYFORWHENHUMANS COUNTGODSSUWAKOMORIYA AFROGXGDALOCATIONINTHEOUTSIDEWORLDIHEARDITHASASEANAZRINEATSTH
 ERICESQUEAKPOSITIVETHINKINGASUSPICIOUSLYMANIPULATIVEWAYOF THINKINGBYSELLINGTHINGSHINGINGONTHELINEDONTYOUKNOW
 WIMDOINGTHISFORYOU

Kunci

Matriks kunci 5×5 hasil rekonstruksi:

S	Y	M	P	O
I	U	L	N	A
T	E	B	C	D
F	G	H	K	Q
R	V	W	X	Z

Susunan ini mengikuti pola pembentukan kunci Playfair pada umumnya: deretan kata kunci SYMPOIULNATE di awal, lalu dilanjutkan sisa alfabet secara berurutan (BCDFGHKQRVWXZ). Keteraturan tersebut menjadi bukti tambahan bahwa kunci yang ditemukan memang kunci asli, bukan sekadar kunci yang kebetulan berskor tinggi.

Referensi

Tabel referensi frekuensi bigram Bahasa Inggris yang digunakan pada Tahap 1 mengacu pada [3, 4]. Tabel QUADGRAM yang dipakai sebagai fungsi objektif pada Tahap 2 disusun dari 45 *quadgram* tersering pada berkas statistik `english_quadgrams.txt` [5], dengan nilai persentase dihitung terhadap total kemunculan seluruh *quadgram* pada berkas tersebut.

2.4 Trinitarian Rhapsody

Deskripsi Soal

Di antara tumpukan lembar catatan risetnya, Marisa menemukan sebuah naskah analisis yang mengulas entitas penguasa tiga alam, sosok yang skalanya melampaui siapa pun di Gensokyo maupun *Lunar Capital*. Naskah yang menyimpan penjelasan mengenai eksistensi mutlak tersebut tidak dapat dibaca karena teksnya telah dienkripsi menggunakan **Affine Cipher**.

Bantulah Marisa mengungkap pesan tersembunyi tersebut tanpa bermodalkan kunci awal maupun *known-plaintext*. Manfaatkan sifat matematis aritmatika modulo pada *Affine Cipher* beserta analisis frekuensi kemunculan karakter dalam bahasa Inggris untuk melakukan **Ciphertext-Only Attack**.

Berkas lampiran:

- Berkas *Ciphertext*

2026/2027

Langkah Penyelesaian

Artefak

Plaintext Hasil Dekripsi

Kunci

Referensi

Bab 3

Problem Set (Bonus)

3.1 Ex Machina

Deskripsi Soal

Di tengah *hollow night*, sebuah transmisi aneh muncul dari salah satu wilayah yang telah lama ditinggalkan. Sinyalnya berulang secara teratur, tetapi tidak seorang pun mengetahui siapa yang mengirimkannya ataupun sudah berapa lama mesin tersebut aktif.

Setelah diteliti, pesan tersebut diketahui telah dienkripsi menggunakan **Enigma Machine M3**. Namun, setiap percobaan untuk mendekripsinya menggunakan konfigurasi Enigma M3 selalu menghasilkan teks yang tidak masuk akal. Entah karena pengaruh *hollow night*, kerusakan pada mekanismenya, atau memang karena mesin tersebut sejak awal dibangun dengan aturan yang berbeda, Enigma ini telah dimodifikasi.

Analisislah perilaku mesin tersebut, **temukan perubahan yang membedakannya dari Enigma M3 standar**, kemudian lakukan **kriptanalisis** untuk merekonstruksi konfigurasi yang digunakan dan memulihkan pesan aslinya sebelum *hollow night* berakhir.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/1TcBU6s876b5ah9EVsuX_xHDTc_0LaWsI/view?usp=sharing
- md5sum: 1074d1cfcaf2482f7229c8f7d40ba097
- sha256sum: fd31f4b0b6475fc15926caea6b28179ea2a1228fe9f063cad8b2d5c215d9031b
- Kode mesin: src/31/enigma-machine/

Berkas *Ciphertext*

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UTJX CENI RYOW WSVU YEHC YRPR BJWZ NXDJ UNVX WXFE
WOUO DQDZ XJTI QKBI USMI AXEI JFZX XQWJ IBXM PCBW
BAFH OAXG BPBD KASZ LQOY ASNA ZPPC UTVF ISQZ HILW
GYCJ AUQV UGXH VQWW OXMX WEBW XSBQ EGZM OSKQ OYWK
MLKZ YPRC MMHW EHEQ DYJU XLVA CJPN SSYH WFPM WKGS
SAVE XVMF ZGCI SPEJ EFHD RXBS LZDX WYYU UWHB WTJG
ZFJM MTUV HHEZ GVKU IRON NNUU VWFD XZGW BUHB KSSG
VBHP KCQN CJCP GILT KZZK FMLW YVNS OXYJ AXQF RNML
NRYI RLIP CYPC WHQI YLUS JDDH HIDZ OWMH HHML CXXG
BJPY HWFI CMGP XGBP LBHZ ESGG XATC ERAY OSFN MKBP
WWKA PXHU UVIG ICOZ PDLW IMYC XZZH YQUV GUPM WYUE
RYPG TMWO GMTC PMXK ZLPS ACWW UFKO SJST OHGW AXOL
JNBX SQUP UIAP OMLU THXE LCGR HVDP HYLU HQUX OJMO
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NCWD WIYK BZXM RPBO SYGI ZZYF AKCQ HVCN OHBT XEGI
GHXE YBZL OHSA DDZF AZPB AQVL AMCI TBFN TBCS ZNQO
FRCD WZRB KJEL MWDC RDAB NQIH DJKQ PYYB GNLN GWWL
CNCZ NEDW FXXR XPTH LVWN ZCXB TNYH PQUG TNFV AXFE
KSGO HNZZ PKHK HYQL SWQL RWXV LVAC WYWN IFBU JEPX
YSNJ IYXB JVNN QMEI FHMY KJTR YBAM HQBF FMWJ ZZVF
WUMS UPVJ BQUQ OJNP QWMM NABE YCCF RJTQ UHGL MMYM
KMIH VDKC EAOJ YCPB CDQV OEIL FLLO MGHG PPPA QOUS
UYIN MKBE HBGW VGVX ALOJ DGRD FBUQ XYJQ AYHV EKZY
SUJG LADI LTQT DXFZ AHLY LZWZ TBRJ BGKZ NRAB ZKXH
YYTB POXJ MXCP HDPR FCUG CSRB TQQY ZTBH ZZKP XEBO
TLZN XLOY ZSSV MQIU NVMP PSOG HBJZ CTPD EGXV AGWQ
MJBZ AMSI DYKS TZGB LPIW RMKO ZRYY OLTO YKKB YZWJ
EDDR RRLY WZRA PKNF IJGF SBJD IOQW BUCF PUEQ MYOZ
DBVQ GYUB CQJX VCSA TQCY XSUL RFNL ODFC KTKD CCIS
AZFE MHPR SAPV CGFQ HCAB JVBK CSWW CERU UOJX FERO
FGYS PIRG MDTP YWWV BOIK LOBZ QQON QCSN YZPM EEZP
NAJC TDLK VMAL VMRT OIHJ YKGL AYGM YCJE RKZD BBKE
QFNC IIEG JXMQ BMAS KJVG GITP HUTX YTMR YGIA YTHP
DRAF OZEG IHJV SARO HTPJ IVRN XGTJ EXJI MWBK CYCH
NOPN SVMM FRZH RAET IYKA GAKC HFVX BUTT JGGK HKWW
VKBY POVG IYIC KOZA ZDPK VXXE TZZR AJMQ RAVC FJIC
QNNR KJLG ZFMW QOQP KGFT DULW DWWP LHKB UIUP QDCE
CGOG IAIC UCWW GESB YSDW FAFX DYAF QGVZ JBVV VRAF
NAYL JQTN XDYU KANE WKZJ FVAI MSGZ PMKV SRAS PRPD
UVXV NXJI VNAY PVQZ WBPQ FXRG AOGP JXII CPFM ZKWP
IHQY KHOK IHGQ KTQH QTMD MWLG YUUG BOVY QBHG VKXF
LDUG LSWX JAQY SAVN PYFR CCLW OPPB OOMV IHQT SCKN
JTSL ZQLC ISKC LPUI NKJF CQZZ NEDW WXYI SBCO KVPY
RHLV CHMA BOCX GZXT TCHA FJGC GFGU LEGB GESE LXUQ
LZZO BIIP NLYI HDEC OSUH JJZX EPZP PCWP XAVT VNWG
AJQU HXCC ATXZ FMKD NFXC CCSX TNON RNRQ KJQB SODI
HOZQ HCUQ WQJQ YCYD BXHK NSQK RUOM NOCE IXRL PQHZ
NKAJ TAGS ZFSI VQHG ULJV PZPJ KEBX ZXXT UCWX MXBR
MEPQ APOG COGY PJVO MSVW CWOK JYKV MNRB DZTO GATI
XXPN ZHLD OFLX LHWY SFJF IFCZ PDYG BHFQ YCRB GCHA
JAGX PNTK RFOL RWRD ZHPD NOGG JJTO PXPQ RWPU JTQY
YUIE GQJD YCYC MLHM FZLS HNFV QCUY FBCE XXGO FJPT
CQPS YHOD AVPR ALGP JPKB FFDH KLGS ZGXS KYUX BWSX
YPSY MYFY NWYO BNJY FDLR UPNV QRMU YWET RWEF PMCS
SKTK KNVO FPWB QBXA ENXM KMOZ SCPJ SDYX JFCV XPRJ
LDFQ DNCV VMXH WFLH CYIJ XHFC CYAD QRQO AGKT OXGL
VGJC ZFQE MTNZ OBZT AMMC RZMP SFEP YNZO LOIA QSPE
QRVQ LTEM DPZS RJBP YJEV WROV HRFO WJPS BENC AUJH
EDWS GFGJ RNYL BBKG KJXW VLXI EKQL LTUB PSNG XJRR
JZXP PLYP QPVS AXEM QSDI MFOF RMKC PMIU BGCI KMHU
VZRL QRAJ AEUP JQMY SUTY SYCQ FWYQ VDXA PXJV KQBS
HYDU QEUR SZUA FWQF OQDT DUFQ VOKW RWAC BFIF XZNM
VPWU ADXS LNCV LBAK OJNZ JMFO XQWL TZDT SGRB OTKG
DVMA XSQB BDNJ PPGB GIUN JGSV YFER AVPS WDYL XVVA
ZLQB GIBL TGJH DZNH OFKM DXMP VYQG GHIC IZXJ FDBA
XJUL WOZZ XUGC TTHW YGKI NSXI ZREE YYWR IKCS ONQE
NJTZ FZWC ATSM MQLA VLNN BCCS ZBKK FEHZ GTYE ZYVC
XNYB RWXG RHQT RDWK FHJP THQA YPEG ZVQW ZXNX WMRO
YZMU HDDC BAWN MATS XAWM GHDJ TSEJ BOAE EVWU YYMD
YJQU NFEC QLWD WTKG CBTN VPLA DQDY HAYB UPLJ ZHTB
XMLB MTUP AMWM IONR MJRB XOYL PYZU JTRY KATU UAOZ
MLBV JZLL BDYV MXYR OQGX GRUK STVT MLSX OLZJ IEJT
NLRV HNUV PIKX VOBQ EXMU BHZK FHNC UNCN WVEY BHRU
XXSW IXFC DAUC ZYLR IECI ZVDW CTUN FVDE RJZP TRQY
EHMF MDDS DJTE BVIO ZFVO YPXM KKXA SYNS BFFW HOLQ
FZVA IFHS PZJB YAYH CMMD OTLD GEVL LLXJ SEKD WFAD
HYYG ZIIV ZOOM WANE XHND CDTU CGAV JOKK MTLE YDDW
QEFP NIVT CCUL DENX SCSU BPUH GIIT TTCN NCQP UBKC
YCXX PWVO TGVS KNXP GDMS DHIG OMQY IVQH IVCG CARZ
HRXX WSRU WWKO EOQC QKQF GRCF KOQD NRUH MNCT BZGP

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JPGH BYXW RMVV QOWW PNCX BFQM BNGV HLXX WFZE WSAY
 SMXY VWOZ LNPV HVHL MSAF EZIJ EVHY EWSI SHUJ VKIG
 SNZI IHGV BIGX DELG NXTU KEWJ LJRG YLES JURS TGMN
 JUBX QSYU BKAQ YMGM BPBI PCMK UYCY YPIS IVDH NNLO
 KNKT MJIE QTKQ IXQF INFF ZOFV NBZS UQLP ZNLF DXCI
 FPRW MIHU RRNL VNFQ DCEB MNZH AEAU FRND JIIP KIEJ
 WPPI ZPON TRZN BNNT DANG VYEJ OGDY SZVG ZEPZ ZOME
 JJJT QHZZ POGO NMYJ DIIZ PQEM YTHF QCFW WTDQ EQML
 CENZ ZOOR FUXJ FLZT YEAQ YOEK QTBQ JBDC DLFV PTDA
 FFFU DYUS MXPV HBGJ EFJZ POMU HMSJ CYVY QOBV XDEW
 LWKV GZXU LHPR WTPV BKMV VESX PFNS PUEN VZCU OEKV
 CHJR YODV METT PSYW WFMS EEZV YGJK DBNA KHER SSXE
 BZVY NLPL FVRV KUBC YOTP TMOT BCXV GGJU YGYH GVBZ
 CLAR HADF VAOX HPGH PVEB XVVY BGTL AMPC UCIL UMCX
 FPCP GHGM MXYV IXVH HVTI MUPU XDCN GISL LLMI JUEH
 RTMV CKFE ULKD NLYP ZZQC VFUI IWZG TGMH LPSI TEJD
 UZTP GBVI XEDU QNCL RIAR UNQA LHAF RFLJ JSTJ BXHL
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 EIGX VGKG NXGL FTEA HEFX BNQU GIKS TJJH CNAJ XQQK
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 PIHB UUMW NHHH RACR LRIR UQWQ NVYS QAQI LWQL RTLR
 XWHD WNSF AURV FPNI DAKE LPJL SZDH RJRV QNEY OTGQ
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 KVXC GNDJ MAQE IXWS SBPI NZST VOUJ EFJC LWVP TGVR
 IBOV DRAP VLDQ AKQB DTJM YNGM GBIY IQZD BBRN KHBB
 ZOEV LFCM FULG FTSU DFIW BSDN RTDF PJWM FKJH FCLR
 PYCN FPYR AJIB SEGV DSGY TZPD CYMH BCLV FXFG DFSU
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 CPUS CFOV ZGKV QSNX ZORB SIGH PXQD WYPY IPQY QTBE
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 PVTI HKPK RXVO ZHBP ZFIM NJZZ BLRY YBJE DLNX RVBG
 ONOK YHLR QENX TSXX PMQP YYAD PYIH PRHA ZRSX WGTW
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 WVRB GJXY ZELC OYWT JXOU VVPM AWSV ZXJU WNSE KQCJ
 EUSP UWTG DLYY QWTM UWAC OFIY BFCY JKXS TSYW EGSG
 DKME FPDQ DLGR UPFN VWUD AYPC UFCP AVIV QMWU XONL
 CGSC EFWR DRUG YQHM SLCT IQOS WRBF WSAM RXWA AIVO
 GCPY SCLZ DWTX RLDZ MLST HHJW XRYD NYMK SIIE DXCD
 YHLC JNLN HNVH YBQW FFFF KSOM VPOR OQUA ODVC NZEM
 QYPS VTAK ZNXX PWOL QYKB JGDP WGHK IRAS HCID LLGC
 VFUW TMUF LOMK NOWC ISNB OUQR ZRVW KUUU BYEO EYDF
 JVDV NGXR BYEE RNJA XMWG IMXR BQFR ZZAK FPNQ GNOI
 OALF VWOK PTRF MFBM PHRL MJAT CHJG CEYD NJVE WJBR
 CGPS LONH XDHN EANP SMYL VJNJ FWVI NWRH LGBB YBHW
 PDGI FWEX BCCC ZPOQ RSAY OUGR QWPX JOHS NAHO EFVG
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 GXNI APKW FNEW HGHP DXEP GPIU CNIB DDIG YCCE TOWB
 OSJZ CHQU ALIO XGUI APYG UDZD QUZZ FBAS EAKX CXSU
 EQDT EUIP SWUT BESM BDFG FQZB SITO ZHEB AIPO QISJ
 BBZS BWHQ CTHR PBKK DRPQ UTTG RXJJ SKVC OJUW MLRR
 IEUK GFTL NUUZ BMIF GYWI BFUK MMXJ ZSTL VFLR FBFR
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 DYFV YDPK YXWP YQFQ SKQA RWGZ JJJJ YJBF VTSU HOVW
 EIRO RIKC MPPF MMQY VDOU QAEJ DHUO PINZ BPQD ENDH
 CDJG THOB DERH RMZJ BKDK PDFF RPQN UQRG KNFC NMPE
 NCKM NPZJ RKGM RKPI WNXS PEGG BRPL HQUC KVJJ UMFM

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KVZR YOGC YJXM UZVN DRUK WBHQ XZPH DPYI LQUV KBPP
TYNF UVZZ YQWO JTSN IFNQ LPLR JZEJ ZZQJ LNKQ OKQT
UMLF QVOZ WSLP THJF DJUC KQEX ISCX QCEP VCQC NMRR
QLXZ XFOP FZVP ZXGJ RQSS BLPS JDZZ PLSW RCPC MCNG
FCJD PQIO GNCA MYHA GOCF BVZE REZY WIDC DRYX GQJW
VLKC KMXZ KTRT SQDH EDKW GDBT KBRX CZRK HDYU HVIF
WNFX QGFT OYQU NQLB SBRQ QFGR FTXF HCCV XQWW CZQN
SZCG VFMZ EZGD XSWI WFZU SPBQ CCOJ GLZI GRUY DREP
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AJKZ CNCR TAMK FXHY ZCAH GHTU ZTKF PCBR HPEK VXCD
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CZJS DRBV JCLT QFNR YQRS FKXT TBWY LDVC DJNJ CVED
TXQB AVIW VVDB WZHY WGQQ ZZYF NVXW OPBQ CRDD BDLW
SCQG CGXD UTMB RBGA VMWJ MFEE QHBM PYDL YMGC IJKF
OUES HWWO ELRO XHHF CLSV QBTT BPTO NGKM DDWB FGAP
DEYU JNHR FQAY DPTA TRND NBCI SBHR OASR CMJU HPXQ
KLDD IVHA SHNT HGPZ PXWU MGYL JWLF DSWE SMQN AQOM
MIQX MIPV PWPA YBMN VWSQ WGJX EPHY ICAB MEMB NMVI
XLUO XOUB SOOY PXIS CBQP ZDNU PDYU JQUA GXHX KSJK
FGUJ YQDW EIXP VBSE IANY MWIR ITTP MUVF SUYV XUAR
YKSO JEVZ AMFS QZPY TKIN JZVO IGFK LNGL MNUK RIEU
CKVQ BQVL VIWT MZKP VKZZ FRBX CJXC ORRD SHMO LWBQ
OSVO VOVM XUIZ CNLJ FWVF QSNB NVXN CHFG XYYO GVLU
EDZK OWBC AWOX XTSI XRQO XSHS NBJP CQKW XPLB BBJT
QLYS NBLB QXSN UXPL KSYF BFUH INXV ZPTN YQTF RQCO
AYDG ZFVB MCAG ETBM XZYJ EVEP BCXU ZVQJ PEBD DZUX
JFNL VMZY CKXS QRIN MZWL KUVY VPEJ UHLR REWR BUGH
LCGI TBPV VEAG ETCM VBDR CFVM AZTS JLPY OJPM JTKB
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JTZV MONB YISX SVEE CHXG JNLY QBYA WPQZ STDO BEJF
JEMV WPFO BJZP LIDL GGOG CPVR JQEP OLNH HVCR QNWQ
LQHJ BXFD F

```

Analisis Modifikasi Mesin

Langkah Penyelesaian

Artefak

Plaintext Hasil Dekripsi

Kunci / Konfigurasi

Hasil *Decrypt* Sesuai Format Sumber Asli

Referensi

Referensi

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